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Lee et al.

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(54) **EMBEDDED DOWNLIGHT AND RADAR SYSTEM**

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H01Q 21/06 (2006.01)
G01S 13/32 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01Q 21/065** (2013.01); **G01S 13/325** (2013.01); **G01S 13/524** (2013.01);
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(58) **Field of Classification Search**
CPC G01S 13/395; G01S 13/325; G01S 13/524; G01S 13/87; G01S 13/931; H01Q 21/065; H01Q 25/008
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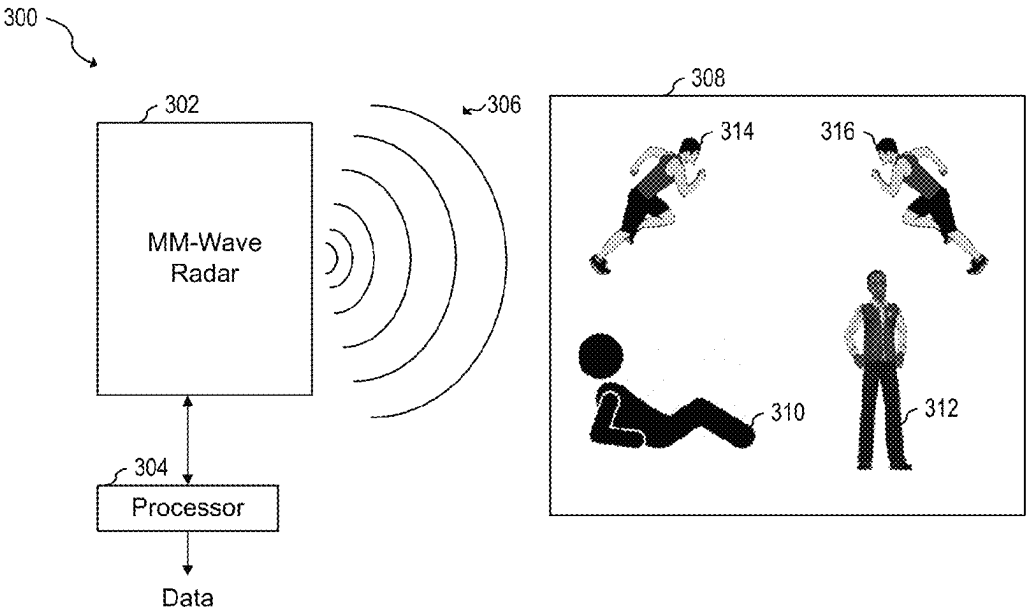
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(74) *Attorney, Agent, or Firm* — Slater Matsil, LLP

(57) **ABSTRACT**

In an embodiment, a downlight includes: a plurality of light emitting diodes (LEDs) disposed in a housing of the downlight, and a millimeter-wave radar. The millimeter-wave radar includes: an antenna disposed in the housing, a controller configured to: detect a presence of a human in a field-of-view of the millimeter-wave radar, determine a direction of movement of the detected human, and produce log data based on the direction of movement of the detected human, and a wireless module configured to transmit the log data to a wireless server.

26 Claims, 14 Drawing Sheets



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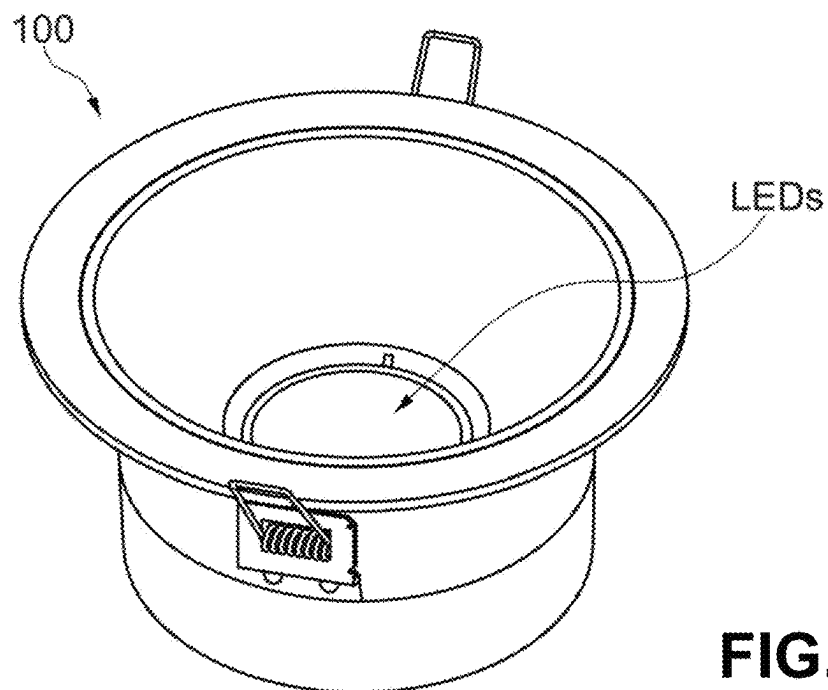
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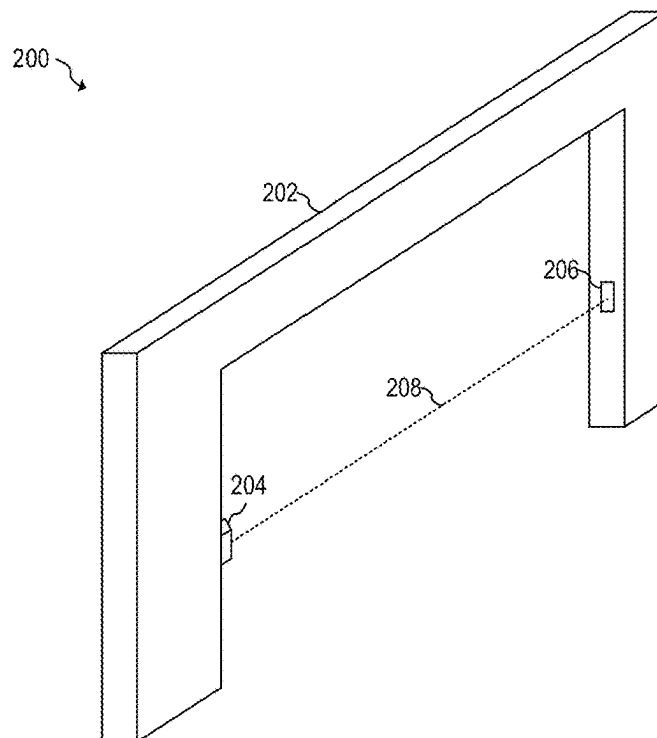
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Prior Art



Prior Art



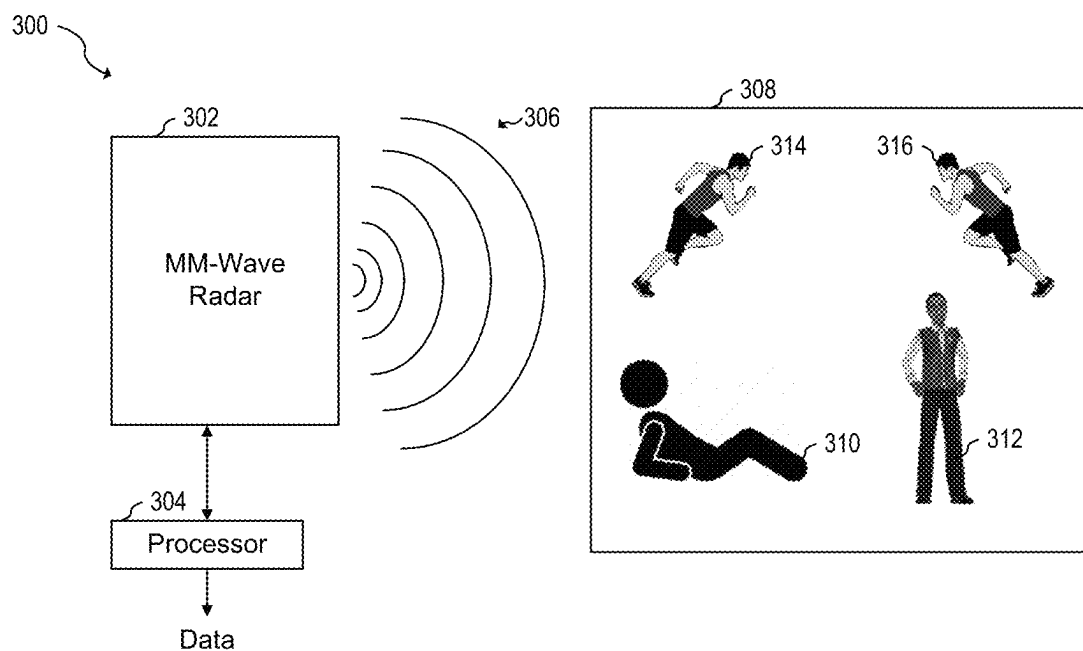


FIG. 3

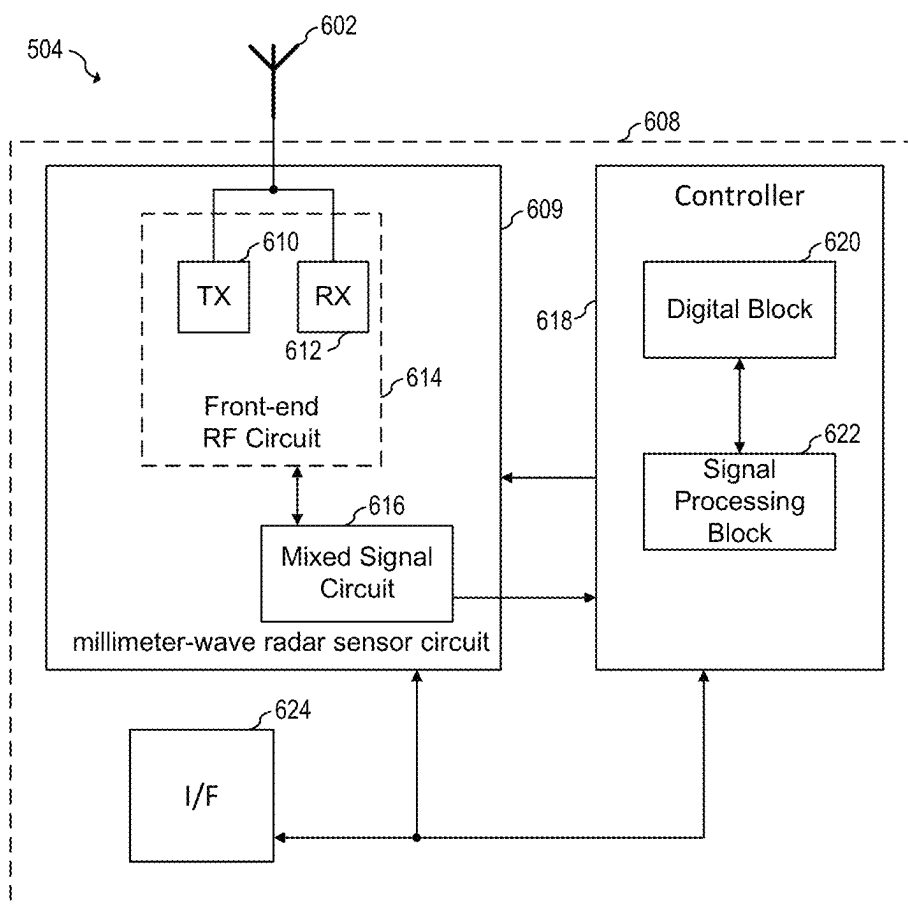


FIG. 6D

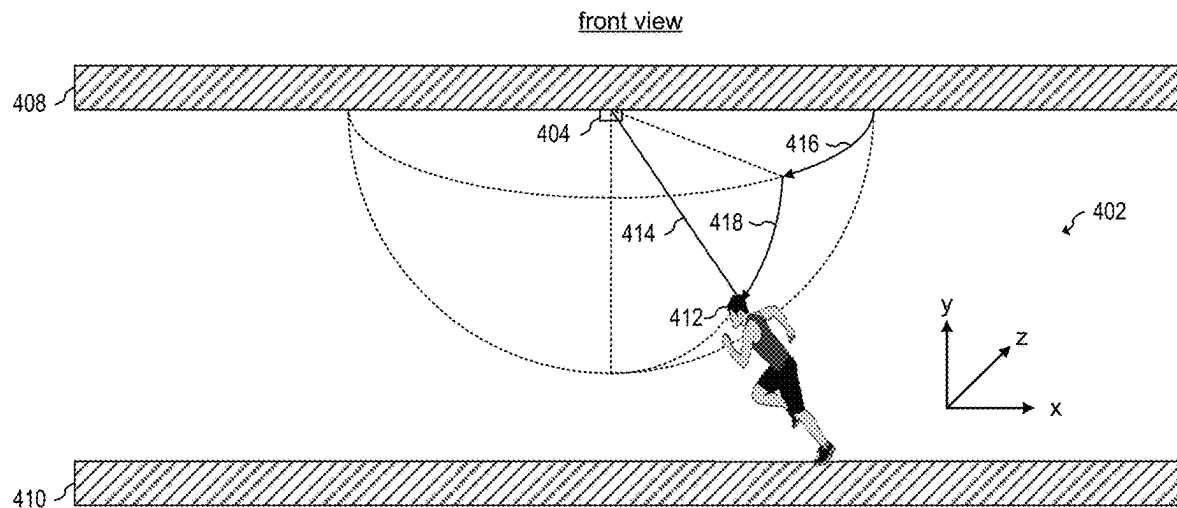


FIG. 4

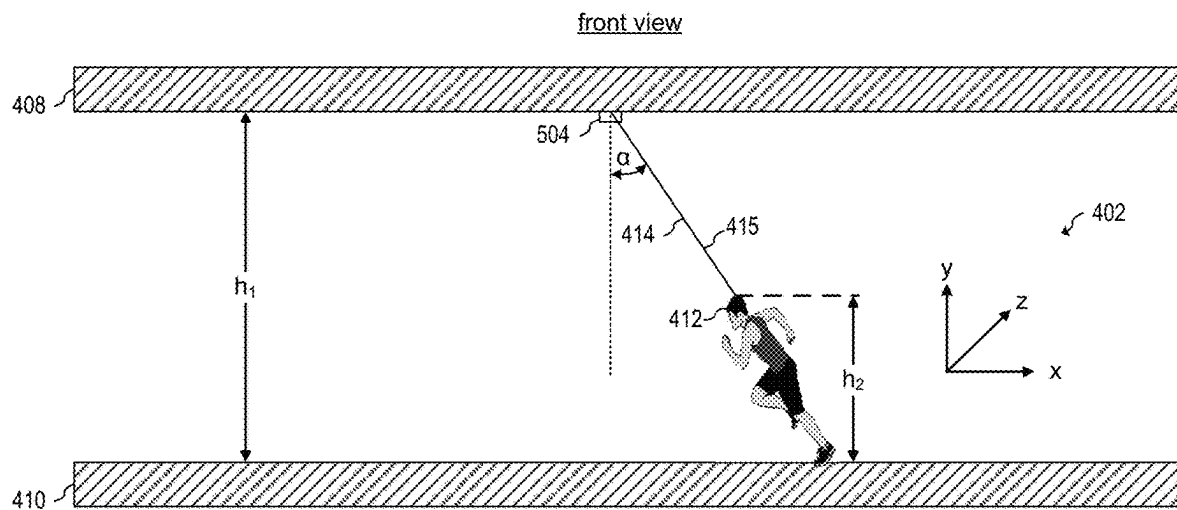
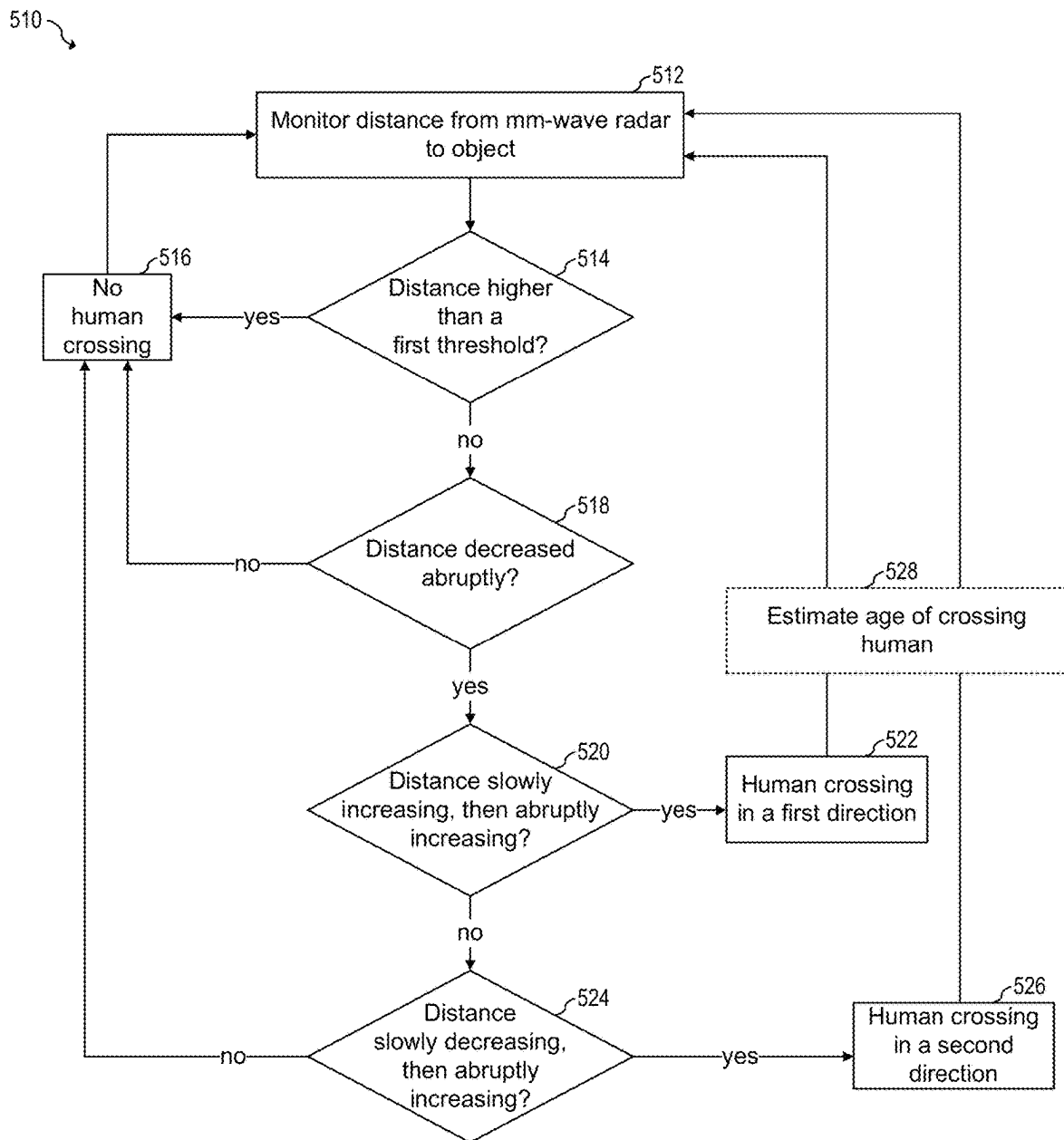
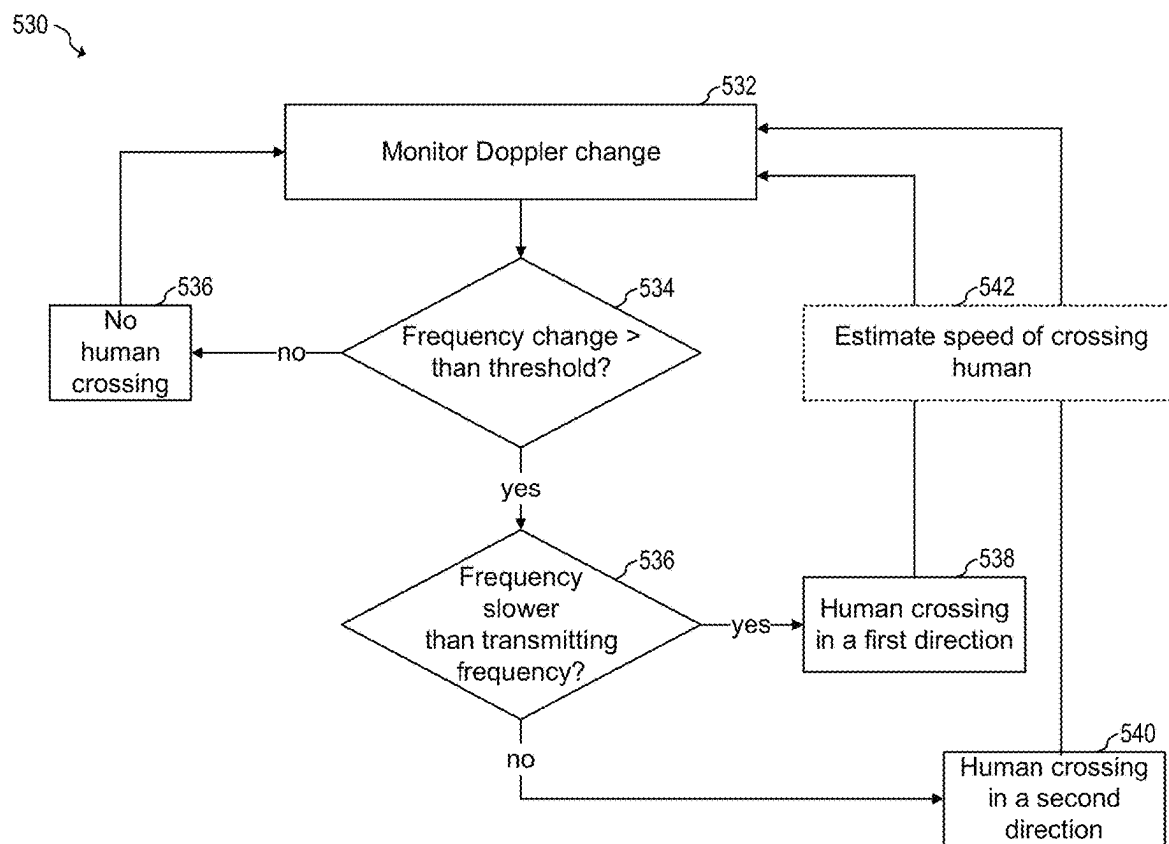
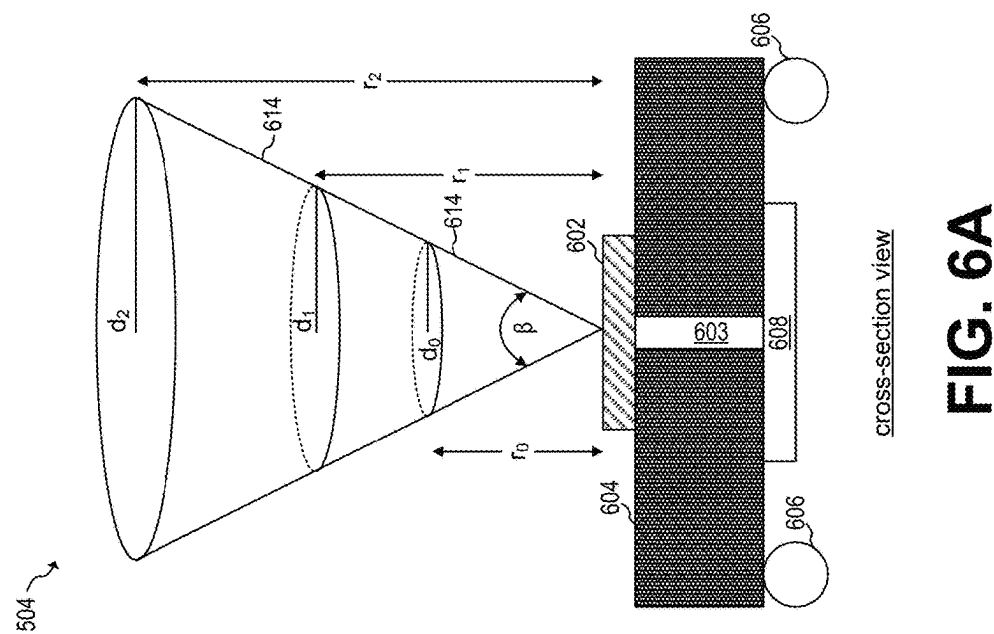
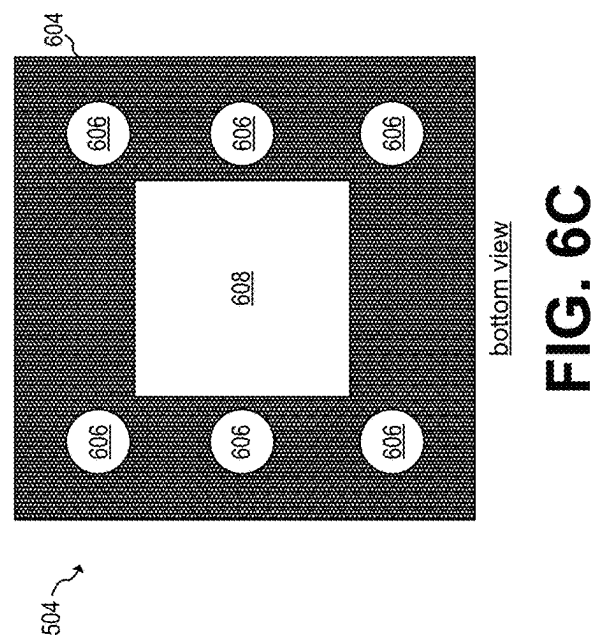
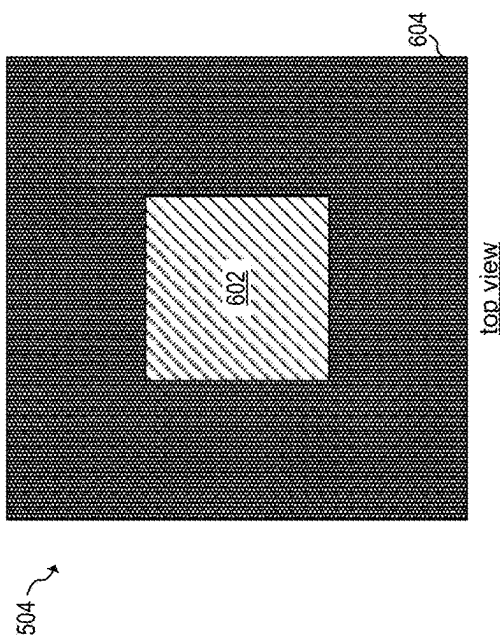


FIG. 5A

**FIG. 5B**

**FIG. 5C**



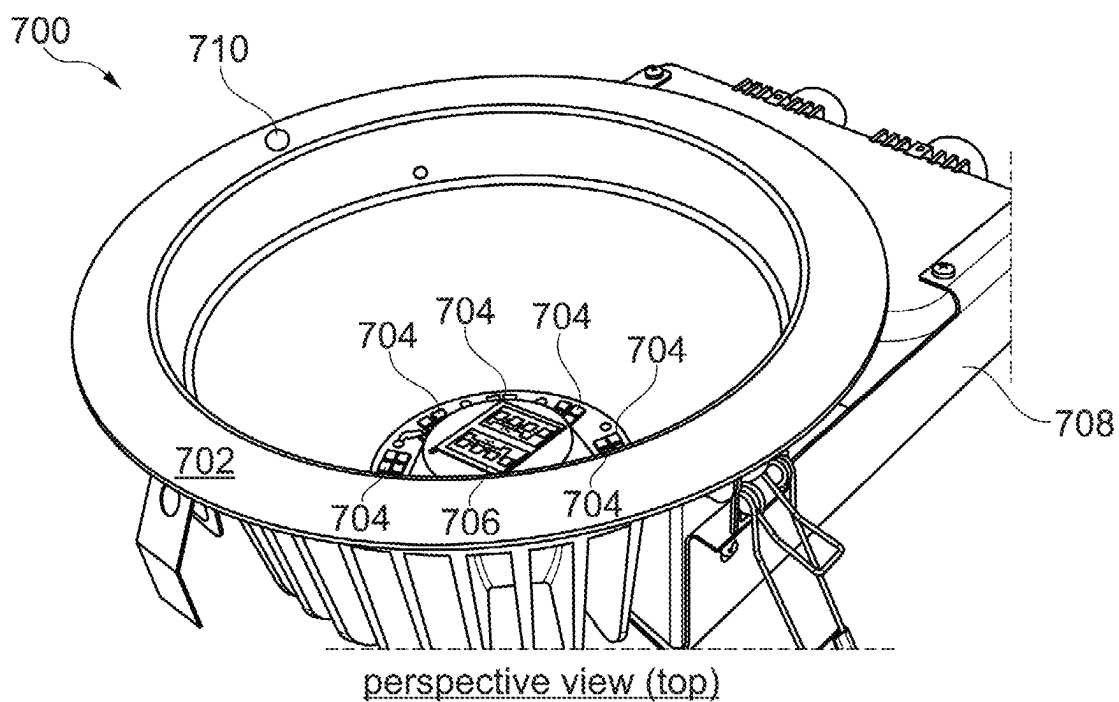


FIG. 7A

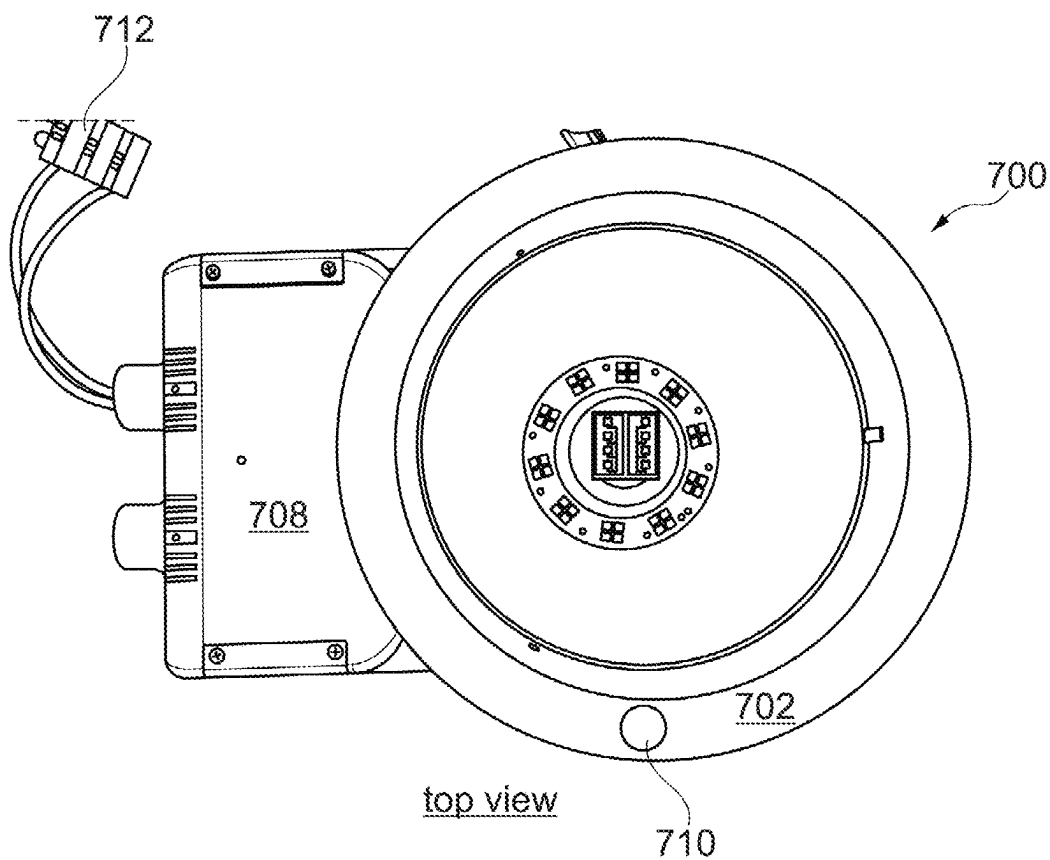
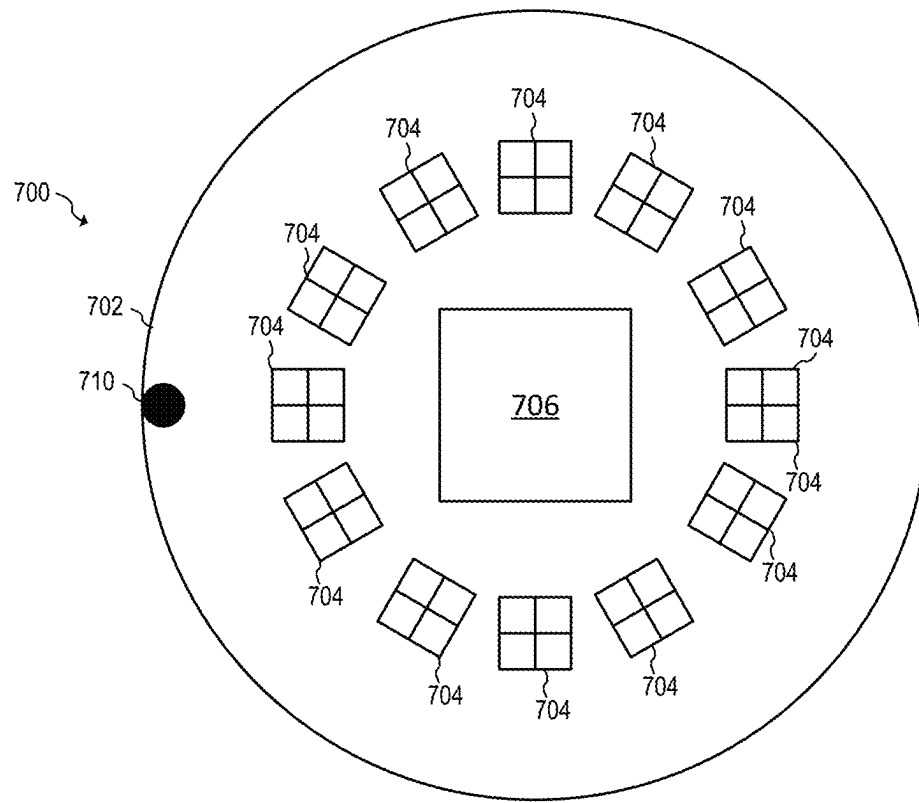
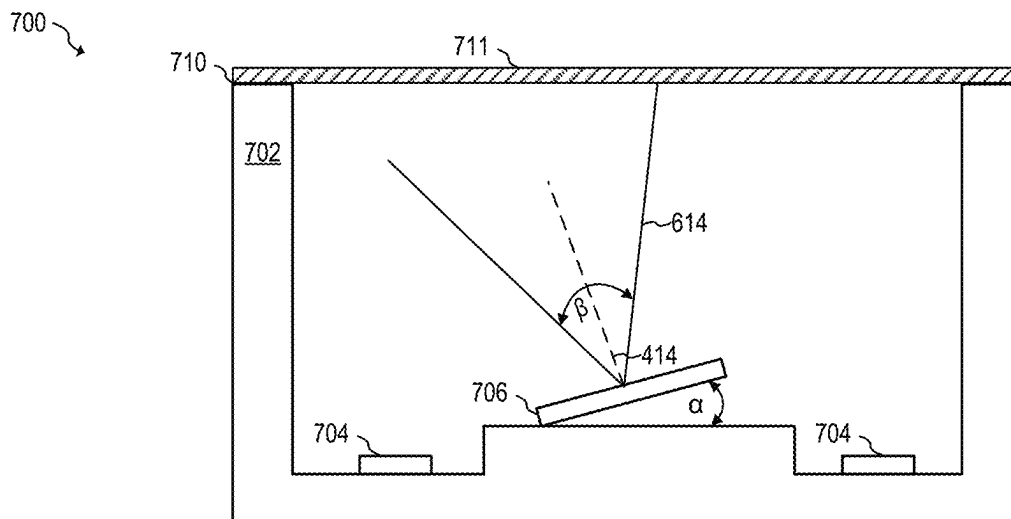


FIG. 7B



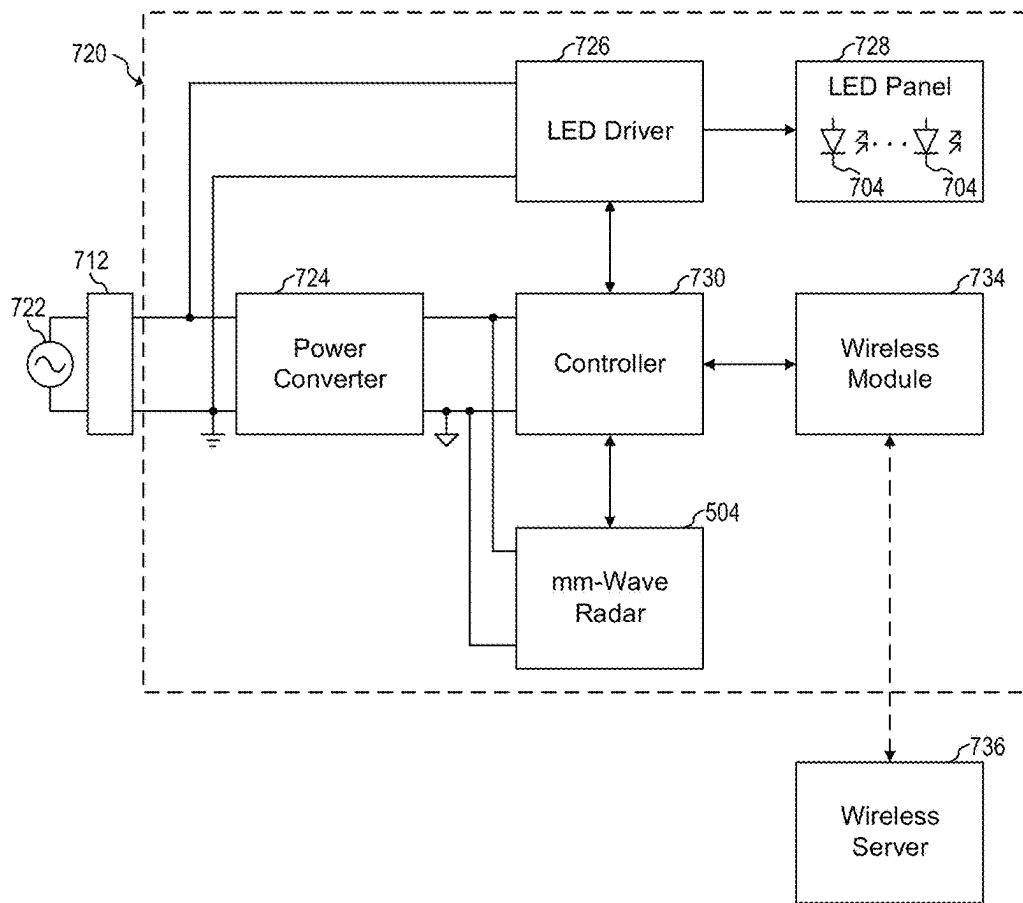
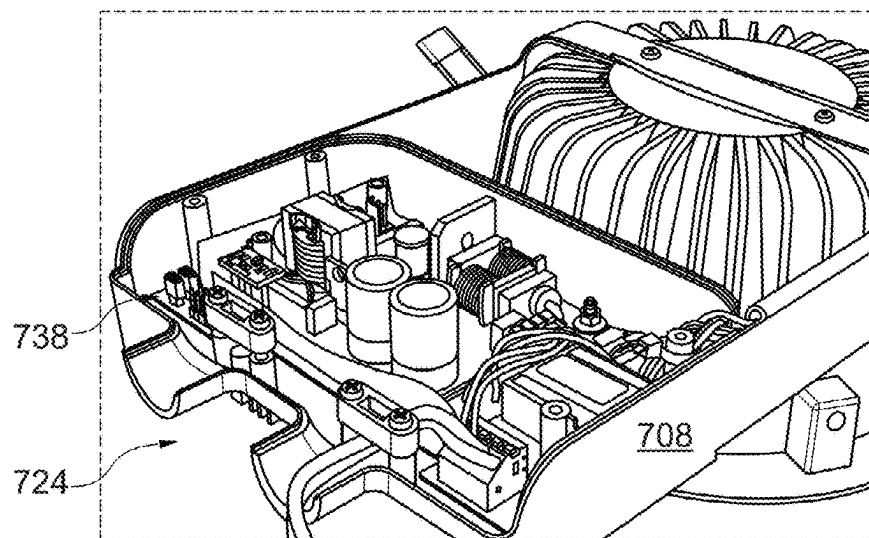
top view

FIG. 7C



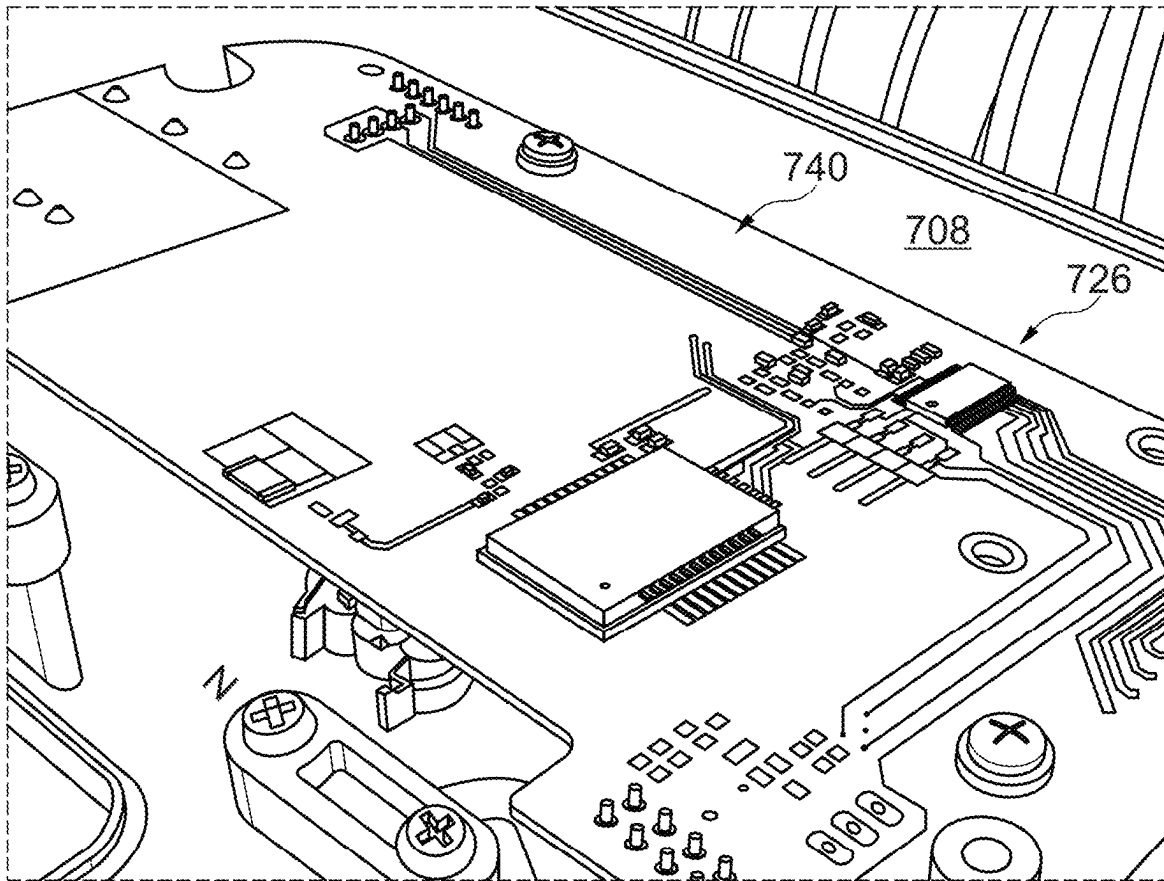
cross-sectional view

FIG. 7D

**FIG. 7E**

perspective view (bottom)

FIG. 7F



perspective view (bottom)

FIG. 7G

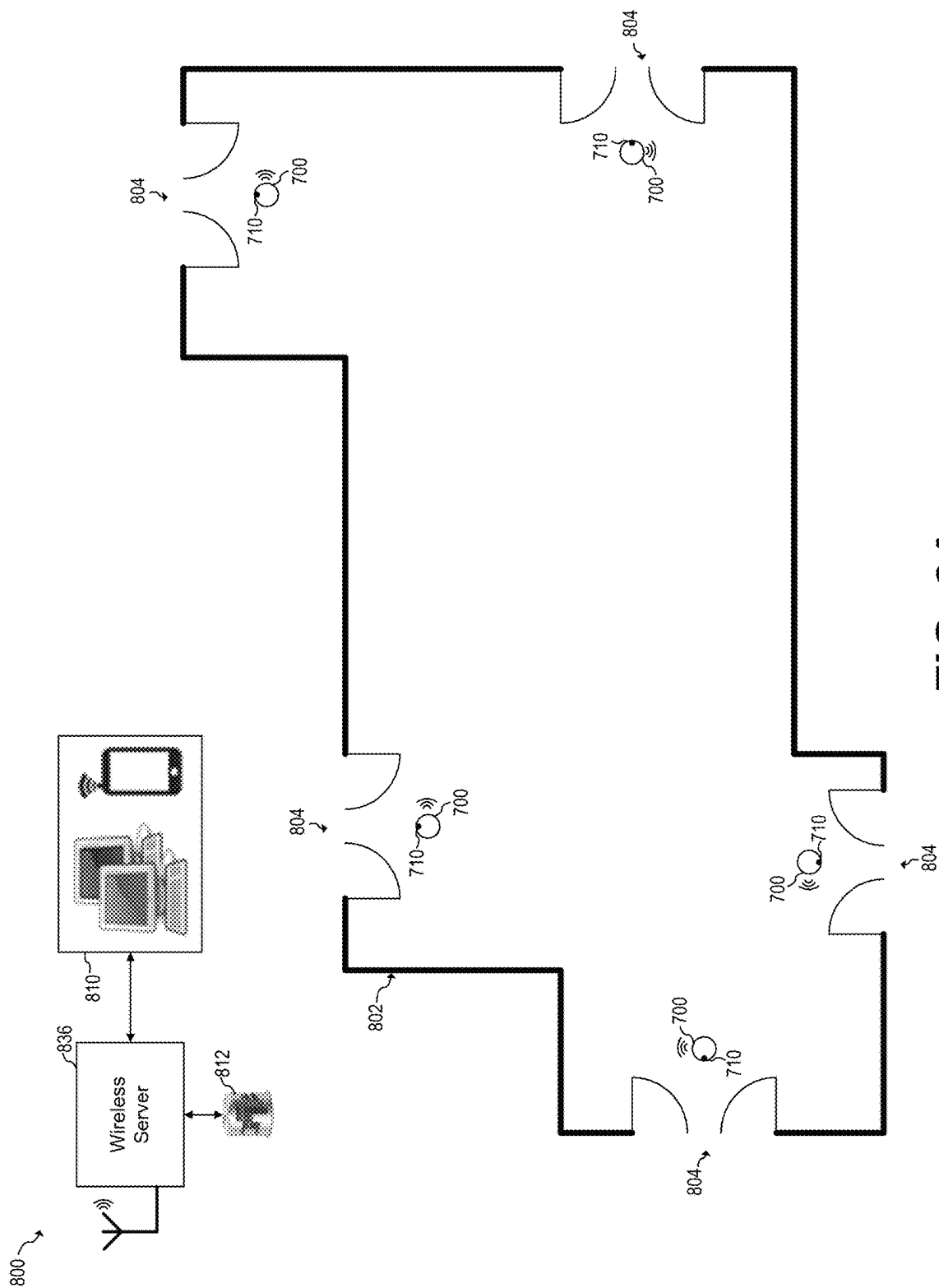
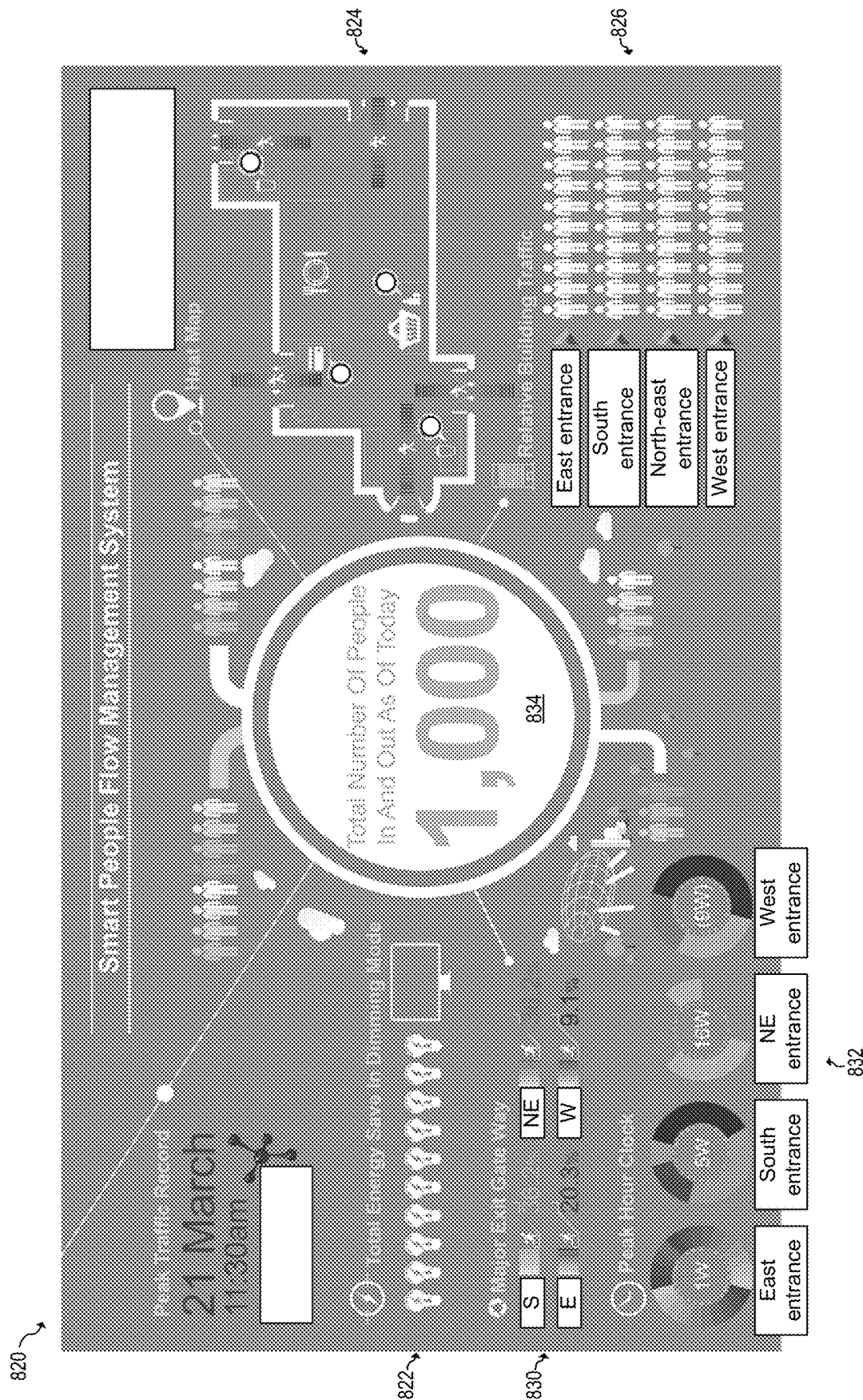
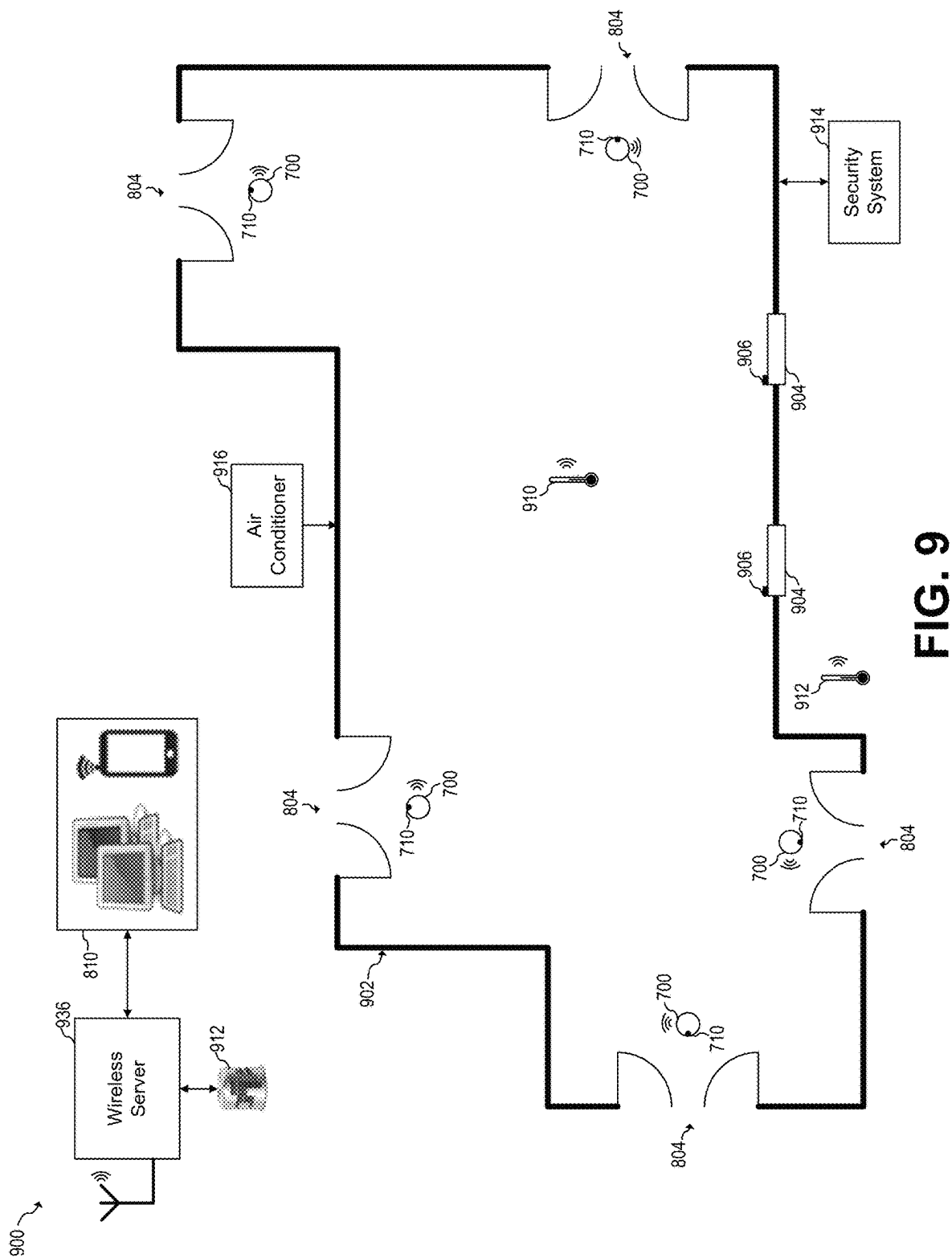


FIG. 8A





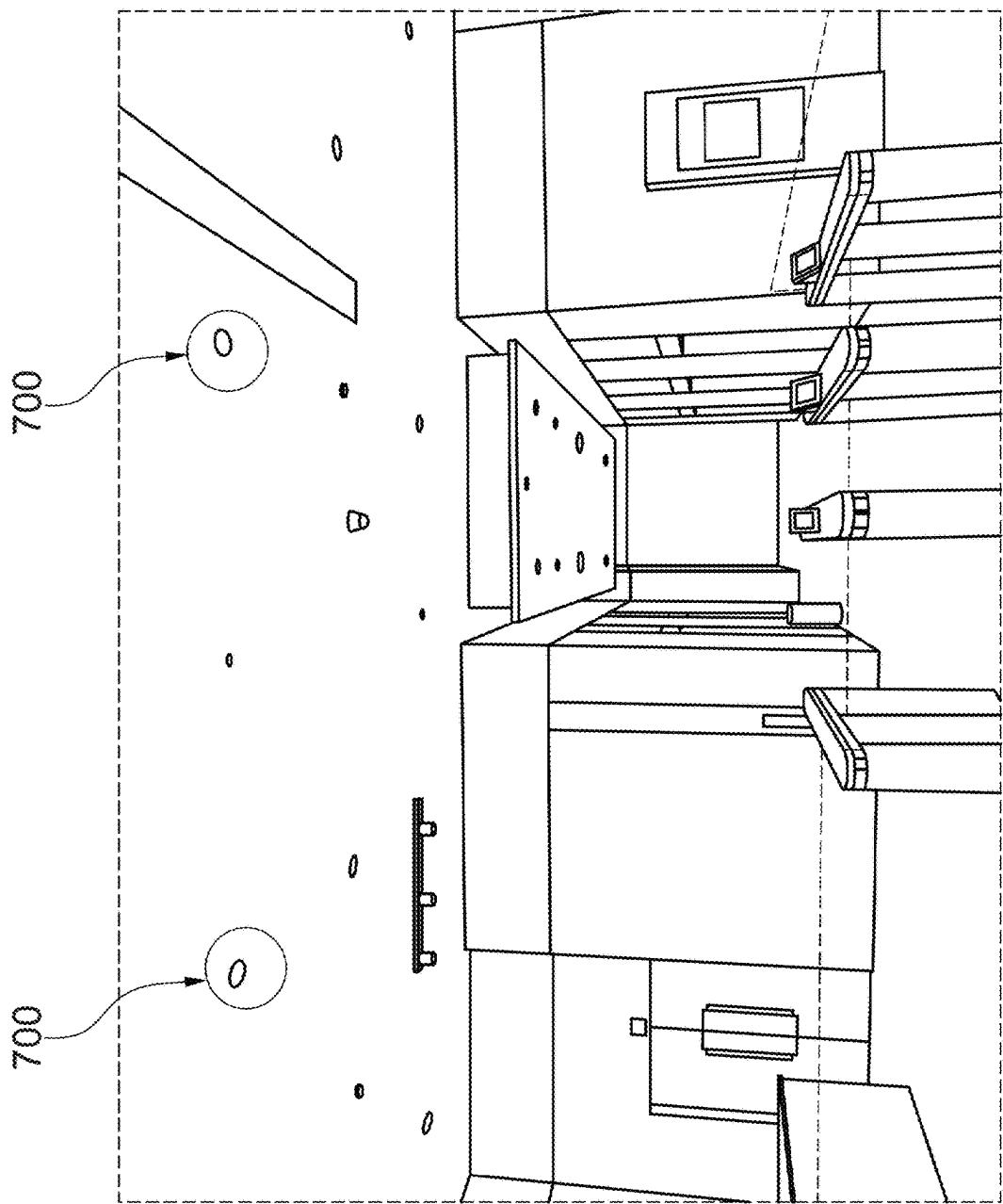


FIG. 10

EMBEDDED DOWNLIGHT AND RADAR SYSTEM

This application is a divisional of U.S. patent application Ser. No. 16/130,507, entitled “Embedded Downlight and Radar System,” and filed on Sep. 13, 2018, which applications is hereby incorporated herein by reference.

This application is related to U.S. patent application Ser. No. 15/950,615, filed Apr. 11, 2018, which applications is hereby incorporated herein by reference.

TECHNICAL FIELD

The present invention relates generally to an electronic system and method, and, in particular embodiments, to an embedded light and radar system.

BACKGROUND

Light bulbs are ubiquitous in modern society. Efficient light bulbs are often desirable. Light emitting diode (LED) light bulbs are light bulbs that use LEDs to produce lights. LED light bulbs are more efficient than conventional incandescent lights, which produce light with a glowing filament heated by an electric current. LED light bulbs are also more efficient than fluorescent lamps. In addition to being more efficient, (or as a result of being more efficient), LED light bulbs produce less heat than conventional light bulbs.

LED light bulbs use LED drivers for driving the LEDs. LED drivers and associated circuitry of a light fixture typically convert AC power (e.g., 60 Hz 120 V), e.g., from the grid, into DC power and deliver DC voltage or a PWM signal to LEDs for generating light. Although some LED light bulbs cannot be dimmed, some LED light bulbs are dimmable by modulating the PWM signal to adjust the light intensity produced by the LEDs.

Some LED light bulbs and light fixtures include LED drivers inside the light fixture. Other LED light bulbs and light fixtures include LED drivers outside the light fixture.

A variety of light fixtures exists. For example, a downlight is a light fixture that is typically installed in a hollow opening in a ceiling, and concentrates the generated light in a downward direction. Some downlights use LED light bulbs to produce light. For example, FIG. 1 shows conventional downlight 100 using LEDs to produce light. Downlights are popular in shopping malls, offices, industrial settings, airports and other settings.

SUMMARY

In accordance with an embodiment, a downlight includes: a plurality of light emitting diodes (LEDs) disposed in a housing of the downlight, and a millimeter-wave radar. The millimeter-wave radar includes: an antenna disposed in the housing, a controller configured to: detect a presence of a human in a field-of-view of the millimeter-wave radar, determine a direction of movement of the detected human, and produce log data based on the direction of movement of the detected human, and a wireless module configured to transmit the log data to a wireless server.

In accordance with an embodiment, a method including: detecting a presence of a human in a field-of-view of a millimeter-wave radar embedded in a housing of a downlight, determining a direction of movement of the detected human, producing log data based on the direction of movement of the detected human, and transmitting the log data to a server using a wireless transmission channel.

In accordance with an embodiment, a method includes: determining whether humans are entering or exiting a building using millimeter-wave radars embedded in respective housings of respective downlights of a plurality of downlights located at entrances of the building, calculating a first number of humans entering the building during a first period of time, calculating a second number of humans exiting the building during the first period of time, and controlling a brightness of light produced by the plurality of downlights based on the first and second number of humans.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIG. 1 shows a conventional downlight using LEDs to produce light;

FIG. 2 shows a conventional indoor infrared system for detecting humans at an entrance;

FIG. 3 shows a radar system, according to an embodiment of the present invention;

FIG. 4 shows a bistatic millimeter-wave radar, according to an embodiment of the present invention;

FIG. 5A shows a monostatic millimeter-wave radar, according to an embodiment of the present invention;

FIG. 5B shows a flow chart of an embodiment method for detecting humans crossing a plane, according to an embodiment of the present invention;

FIG. 5C shows a flow chart of an embodiment method for detecting humans crossing a plane, according to an embodiment of the present invention;

FIG. 6A shows a cross-section view of the monostatic millimeter-wave radar of FIG. 5A, according to an embodiment of the present invention;

FIGS. 6B and 6C show a top view and a bottom view, respectively, of the monostatic millimeter-wave radar of FIG. 6A, according to an embodiment of the present invention;

FIG. 6D shows a schematic diagram of the monostatic millimeter-wave radar of FIG. 6A, according to an embodiment of the present invention;

FIGS. 7A and 7B show a perspective view and a top view, respectively, of a LED downlight with an embedded millimeter-wave radar, according to an embodiment of the present invention;

FIGS. 7C and 7D show schematic diagrams of a top view and a cross-sectional view, respectively, of the LED downlight of FIGS. 7A and 7B, according to an embodiment of the present invention;

FIG. 7E shows an electrical schematic of a circuit of the LED downlight of FIGS. 7A-7D, according to an embodiment of the present invention.

FIGS. 7F and 7G show a perspective view of portions of the circuitry inside the enclosure of the LED downlight of FIGS. 7A-7E, according to an embodiment of the present invention; and

FIG. 8A shows a top view of a schematic diagram of a building that includes a plurality of the LED downlights of FIGS. 7A-7G, according to an embodiment of the present invention;

FIG. 8B shows a graphical user interface (GUI) that may be used to access the wireless server of FIG. 8A, according to an embodiment of the present invention;

FIG. 9 shows a top view of a schematic diagram of a building that includes the plurality of the LED downlights of FIGS. 7A-7G, according to another embodiment of the present invention; and

FIG. 10 shows LED downlights of FIGS. 7A-7G installed on a ceiling, according to an embodiment of the present invention.

Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale. To more clearly illustrate certain embodiments, a letter indicating variations of the same structure, material, or process step may follow a figure number.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

The making and using of the embodiments disclosed are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

The description below illustrates the various specific details to provide an in-depth understanding of several example embodiments according to the description. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials and the like. In other cases, known structures, materials or operations are not shown or described in detail so as not to obscure the different aspects of the embodiments. References to “an embodiment” in this description indicate that a particular configuration, structure or feature described in relation to the embodiment is included in at least one embodiment. Consequently, phrases such as “in one embodiment” that may appear at different points of the present description do not necessarily refer exactly to the same embodiment. Furthermore, specific formations, structures or features may be combined in any appropriate manner in one or more embodiments.

The present invention will be described with respect to embodiments in a specific context, a LED downlight having a millimeter-wave radar and implemented in a building (such as a shopping mall). Some embodiments may be used in other types of downlights, (such as a non-LED based downlight) as well as in street lights and in other types of lamps.

In an embodiment of the present invention, a LED downlight counts a number of humans entering and exiting an entrance of a building by using a millimeter-wave radar embedded in the LED downlight. Direction of humans (e.g., entering or exiting the entrance) is determined by using ranging information, such as by using range FFT. Data (e.g., human detection log or raw data) is transmitted from one or more LED downlights to a wireless server using a low-power wide-area network (LPWAN) protocol. The wireless server makes available the data to various electronic devices (e.g., via the Internet). The wireless server also controls the LED downlights based on information received from the LED downlights.

In some embodiments, the wireless server also interacts with additional sensors (e.g., temperature sensors, pressure sensors, etc.) and additional systems (e.g., air conditioner, security system, etc.) of the building. For example, in some

embodiments, the wireless server receives data from such sensors and from the LED downlights and controls such systems (e.g., air conditioner, security system, etc.) based on data received from the sensors and from the LED downlights. In some embodiments, the wireless server receives data from such sensors and from the LED downlights and controls the LED downlights based on data received from the sensors and from the LED downlights.

Some embodiments perform data mining and data processing (e.g., using artificial intelligence and machine learning algorithms) of data gathered by the LED downlights, and by the additional sensors to predict behavior (e.g., number of humans inside a building) and adjusts the LED downlights and other systems based such predictions.

Monitoring people traffic is often desirable in many applications, from stores and malls to factories and airports. For example, FIG. 2 shows conventional indoor infrared system 200 for detecting humans at entrance 202. In particular infrared system 200 includes infrared transmitter (TX) 204 and infrared receiver (RX) 206.

During normal operation, infrared TX 204 transmits infrared signal 208 towards infrared RX 106. When a human crosses entrance 202 (e.g., of a store or shopping mall), the human blocks infrared signal 208, thereby causing infrared RX 206 to stop receiving infrared signal 208. Infrared system 200 interprets the interruption of infrared signal 208 as a human entering or exiting entrance 202. Such event may trigger an action, such as ringing a bell.

Although system 200 detects humans as they cross entrance 202, system 200 does not detect the direction in which the human is traveling (e.g., entering or exiting the store or shopping mall entrance). Detecting the direction in which a human is traveling is often desirable. For example, a landlord of a shopping mall may determine the number of people inside the mall by counting the number of people entering the mall and subtracting the number of people exiting the mall. Such information may be useful for a variety of purposes, including safety purposes, improving the mall experience, and creating marketing materials.

A conventional camera-based system may be used to perform people counting. For example, a camera-based system may use video cameras to capture images of humans entering the shopping mall. Each video-camera may send such images to a centralized location (e.g., a server) using WiFi. The server may then use facial recognition algorithms to count the number of people entering the shopping mall.

Embodiments of the present invention detect humans (e.g., moving, standing, lying down, etc.) and humans' moving direction (e.g., entering or exiting a building) by using a millimeter-wave radar together with signal processing techniques, such as Doppler processing, frequency modulated continuous wave (FMCW) processing, or frequency shift keying (FSK). For example, FIG. 3 shows radar system 300, according to an embodiment of the present invention. Radar system 300 includes millimeter-wave radar 302 and processor 304.

During normal operation, millimeter-wave radar 302 transmits a plurality of radiation pulses 306, such as chirps, to scene 308. The transmitted radiation pulses 306 are reflected in objects of scene 308. The reflected radiation pulses (not shown in FIG. 3), which are also referred to as the echo signal, are detected by millimeter-wave radar 302 and processed by processor 304 to, for example, detect humans and humans' moving direction.

The objects in scene 308 may include static humans, such as lying human 310, humans exhibiting low and infrequent motions, such as standing human 312, and moving humans,

such as running or walking human **314** and **316**. The objects in scene **308** may also include static objects, such as furniture and periodic movement equipment (not shown). Other objects may also be present in scene **308**.

Processor **304** analyses the echo data to detect humans and humans' moving direction using signal processing techniques.

Processor **304** may be implemented as a general purpose processor, controller or digital signal processor (DSP). In some embodiments, processor **304** may be implemented as a custom application specific integrated circuit (ASIC). In some embodiments, processor **304** includes a plurality of processors, each having one or more processing cores. In other embodiments, processor **304** includes a single processor having one or more processing cores.

Millimeter-wave radar **302** includes a millimeter-wave radar sensor circuit and an antenna(s). The millimeter-wave radar transmits and receives signals in the 20 GHz to 122 GHz range. Alternatively, frequencies outside of this range may also be used. Millimeter-wave radar **302** may be implemented, for example, in monostatic or bistatic configurations.

Implementing millimeter-wave radar **302** in, e.g., bistatic configuration, has the advantage of enabling radar system **300** to detect and track the location of a human as the human moves across the field-of-view of millimeter-wave radar **302**. For example, FIG. **4** shows bistatic millimeter-wave radar **404**, according to an embodiment of the present invention. As shown in FIG. **4**, bistatic millimeter-wave radar **404** is located in ceiling **408**. Human **412** is located in scene **402** above floor **410** and within the field-of-view of bistatic millimeter-wave radar **404**. Bistatic millimeter-wave radar **404** may report and track location information of human **412** by using elevation component **418**, azimuth component **416**, and range component **415**. Bistatic millimeter-wave radar **404**, therefore, is advantageously capable of determining the direction in which human **412** is moving, whether human **412** is entering or exiting an entrance (e.g., if bistatic millimeter-wave radar **404** is located at an entrance), and counting how many humans have entered and/or exited the entrance. Bistatic millimeter-wave radar **404** is also advantageously capable of determining the speed of movement of humans, whether a human is lying down, or standing still, and other parameters associated with the physical characteristics and movements of humans.

Implementing millimeter-wave radar **302** in, e.g., monostatic configuration, has the advantage of enabling radar system **300** to detect a human as the human moves across the field-of-view of millimeter-wave radar **302** and to determine whether the human is entering or exiting an entrance (e.g., if the radar is located at an entrance) by using range information. For example, FIG. **5A** shows monostatic millimeter-wave radar **504**, according to an embodiment of the present invention. As shown in FIG. **5A**, monostatic millimeter-wave radar **504** is located in ceiling **408**. Human **412** is located in scene **402** above floor **410** and within the field-of-view of monostatic millimeter-wave radar **504**.

Monostatic millimeter-wave radar **504** detects human **412** by using range component **415**. For example, if angle α is 0° (where α is an angle between centerline **414** of the field-of-view of monostatic millimeter-wave radar **504** and a vertical axis y), range component **415** is equal to height h_1 when no object is between monostatic millimeter-wave radar **504** and floor **410**, and is equal to height h_2 (the height of human **412**) when human **412** crosses the field-of-view of monostatic millimeter-wave radar **504**. Therefore, by detecting the change in height (from h_1 to h_1-h_2 and back to h_1),

it is possible to detect that human **412** crossed the field-of-view of monostatic millimeter-wave radar.

If angle α is greater than 0° (e.g., between 5° and 15° , such as 10°), it is possible to determine the direction in which human **412** is moving. For example, when no object is between monostatic millimeter-wave radar **504** and floor **410**, range component **415** is equal to $h_1 \cdot \cos(\alpha)$. When human **412** is moving in the x direction from right to left (as illustrated in FIG. **5A**), range component **415** may decrease abruptly when human **412** initially enters the field-of-view of monostatic millimeter-wave radar **504**, and then gradually decreases as human **412** moves from right to left. Range component **415** returns to $h_1 \cdot \cos(\alpha)$ when human **412** is no longer in the field-of-view of monostatic millimeter-wave radar **504**. In some embodiments, movement in the x direction from right to left may correspond to human **412** entering the building. Some embodiments may use the phase angle between the transmitted and received radar signals to determine the range component.

When human **412** is moving in the x direction from left to right, range component **415**, initially at $h_1 \cdot \cos(\alpha)$, may decrease abruptly when human **412** initially enters the field-of-view of monostatic millimeter-wave radar **504**, and then gradually increases as human **412** moves from left to right. Range component **415** returns to $h_1 \cdot \cos(\alpha)$ when human **412** is no longer in the field-of-view of monostatic millimeter-wave radar **504**.

FIG. **5B** shows a flow chart of embodiment method **510** for detecting humans crossing a plane, according to an embodiment of the present invention. The explanation that follows assumes that method **510** is implemented by monostatic millimeter-wave radar **504**. However, other millimeter-wave radars, such as millimeter-wave radar **404**, may implement method **510**.

During step **512**, the millimeter-wave radar monitors the distance towards objects in its field-of-view. If no object is crossing the field-of-view (where the field-of-view defines a plane dividing two sides, such as left and right in FIG. **5A**), the distance between the millimeter-wave radar and the closest object is equal to the distance to the floor. If it is determined during step **514** that the distance between the millimeter-wave radar to the closest object is similar to the distance to the floor (e.g., higher than a first threshold, such as 90% of the distance to the floor), then, it is determined during step **516** that no human is crossing the plane and the millimeter-wave radar continues monitoring the distance during step **512**.

If it is determined during step **514** that the distance between the millimeter-wave radar and the closest object is lower than the distance to the floor (e.g., lower than the first threshold), then, the abruptness of the change in distance between the millimeter-wave radar and the object is determined during step **518**. If the change in distance between the millimeter-wave radar and the closest object is determined to be abrupt during step **518**, then if the distance between the millimeter-wave radar and the closest object is determined during step **520** to slowly increase and then abruptly return to the distance to the floor, then it is determined that a human crossed the plane in a first direction (e.g., from left to right in FIG. **5A**) during step **522**. If not, if the distance between the millimeter-wave radar and the closest object is determined during step **524** to slowly decrease and then abruptly return to the distance to the floor, then it is determined that a human crossed the plane in a second direction (e.g., from right to left in FIG. **5A**) during step **524**. If not, then the millimeter-wave radar determines that no human is crossing during step **516**.

An abrupt change in distance includes, for example, a change of the distance between the millimeter-wave radar and the detected object from the distance between the millimeter-wave radar to the floor and a distance between the millimeter-wave radar to the floor minus, e.g., 70% of a typical height of a human. A typical height of a human that is walking may range, for example, between 4.5 feet and 6.5 feet. In some embodiments, the age of the human crossing may be estimated based on the change in distance from the millimeter-wave radar to the detected objects during step 528 since the change in distance is related to the height of the detected human, and the height of the detected human is correlated with age.

In some embodiments, movement in the x direction from right to left may correspond to human 412 entering the entrance and movement in the x direction from left to right corresponds to human 412 exiting the entrance. In other embodiments, movement in the x direction from right to left may correspond to human 412 exiting the entrance and movement in the x direction from left to right corresponds to human 412 entering the entrance. It is understood that the movement of human 412 may include a z component (i.e., movement in the z-axis) without affecting the operation of monostatic millimeter-wave radar 504. For example, centerline 414 may define a plane dividing a left portion from a right portion of scene 402 (with respect to the x direction). Monostatic millimeter-wave radar 504 may detect human 412 crossing from right to left or left to right while tolerating non-zero movements of human 412 in the y axis or z axis. Some embodiments may use a millimeter-wave radar having, e.g., one transmitter antenna and two receiver antennas to improve resolution of the human movement direction.

Monostatic millimeter-wave radar 504, therefore, is advantageously capable of determining the direction in which human 412 is moving, whether human 412 is entering (incoming) or exiting (outgoing) a location such as an entrance (e.g., if monostatic millimeter-wave radar 504 is located at an entrance), and counting how many humans have entered and/or exited the location (e.g., entrance) based (e.g., solely) on range information. By determining direction of movement based on range information (e.g., using range FFT), it is possible to determine how many humans have entered or exited an entrance without performing complex computations (e.g., determining the three-dimensional location of a human in a field-of-view or performing facial recognition image processing).

FIG. 5C shows a flow chart of embodiment method 530 for detecting humans crossing a plane, according to an embodiment of the present invention. The explanation that follows assumes that method 530 is implemented by monostatic millimeter-wave radar 504. However, other millimeter-wave radars, such as millimeter-wave radar 404, may implement method 510.

During step 532, the millimeter-wave radar transmit radar signals (e.g., chirps) towards its field-of-view, receives the echo signals and determines the change in frequency between the sent and received radar signals using Doppler signal processing. The change in frequency between sent and received radar signals is determined during step 534. In some embodiments, the change in frequency is determined during step 534 by measuring the phase difference between the I signal and the Q signal coming out of a mixer circuit that receives the echo signal.

If the change in frequency is lower than a predetermined threshold (e.g., no frequency change between sent and received radar signals), then, it is determined that no human

is crossing the plane during step 536 and millimeter-wave radar continues monitoring the Doppler change in step 532.

If the change in frequency is higher than the predetermined threshold, then, if it is determined during step 536 that the frequency of the echo signal is lower than the transmitted radar signals, then, it is determined that a human crossed the plane in a first direction (e.g., from left to right in FIG. 5A) during step 538. If it is determined during step 536 that the frequency of the echo signal is higher than the transmitted radar signals, then, it is determined that a human crossed the plane in a second direction (e.g., from right to left in FIG. 5A) during step 540.

In some embodiments, the speed of the human crossing the plane may be determined during step 542 based on the magnitude of change in frequency, which is determined, for example, in step 534.

In some embodiments, method 530 may advantageously be implemented using a processing complexity that is low enough to be implemented locally inside millimeter-wave radar 504.

Some embodiments may combine methods 510 and 530 to, e.g., increase accuracy of human detection or other moving parameters. For example, some embodiments that combine aspects of methods 510 and 530 may adjust the threshold for determining what constitutes an abrupt change (step 518) based on the magnitude of the frequency change (step 534). As another non-limiting example, some embodiments may estimate age of human crossing (step 528) and speed of humans crossing (step 542).

As described for example in FIGS. 5B and 5C, having an angle α greater than 0° advantageously allows for determining whether a human is crossing a plane and the direction of the human crossing with a processing complexity that is lower than the processing complexity used for calculating the 3D location of the human when crossing the field-of-view of the millimeter-wave radar.

Monostatic millimeter-wave radar 504 may transmit the raw data (e.g., range/time information) or processed data (e.g., a log including, e.g., that a human(s) entered/exited at time t_0 , or, e.g., a number of humans entered and/or exited during a period of time from time t_0 to time t_1) to a central location. In some embodiments, since the amount of data transmitted by monostatic millimeter-wave radar 504 to the central location is relatively small (e.g., the log data), it is possible to use an LPWAN protocol, such as SigFox, NB-IoT or LoRa, to transmit the data from monostatic millimeter-wave radar 504 to the central location.

Monostatic millimeter-wave radar 504 may be calibrated to optimize operation for a particular height h_1 . For example, some embodiments adjust the transmission power of the transmitter such that it has enough power to detect humans at the installed height while avoiding saturation of the receiver that receives the echo signal. For example, a first test signal may be transmitted towards the field-of-view without humans crossing (e.g., empty floor) to determine optimal strength. For example, the first test signal may have chirps transmitted at different strengths. The echo signal may be analyzed to pick a strength that does not saturate the receiver while still using optimal dynamic range.

In some embodiments, a second test signal (which may be equal to the first test signal) may be used when a test human is crossing the plane to determine the threshold for detection. For example, humans that are farther from the millimeter-wave radar tend to produce a weaker reflected signal. A detection threshold based on the strength of the received signal may be adjusted to allow for identifying humans from the noise.

Calibrating the millimeter-wave radar to be optimized for a particular height advantageously improve signal-to-noise ratio (SNR), which helps reduce the number of false positives and false negative human detection.

In some embodiments, a calibration procedure may be launched remotely (e.g., through LPWAN). In some embodiments, the signal processing associated with selecting optimal parameters may be performed locally. For example, the test signals and calibration steps may be pre-installed in the millimeter-wave radar.

In some embodiments, the change in frequency is determined during step 534 by measuring the phase difference between the I signal and the Q signal coming out of a mixer circuit that receives the echo signal.

In some embodiments, a single monostatic millimeter-wave radar design is advantageously suitable for operation in different areas, e.g., different entrances having ceilings with different heights, entrances of rooms or stores inside a building, locations inside a building (e.g., not entrances), etc.

In some embodiments, monostatic millimeter-wave radar 504 may be calibrated dynamically. In other words, calibration may be triggered by a command transmitted wirelessly during normal operation.

Advantages of some embodiments include extended range of human detection compared to conventional systems. For example, in some embodiments, a millimeter-wave radar installed in a ceiling and having a field-of-view with an angle α of, e.g., between 5° and 15°, such as 10°, is capable of detecting humans and human movements (e.g., direction of movement) with a range of 10 meters or more in the x axis, compared with conventional systems, such as passive infrared systems or ultrasound-based systems, which have a typical coverage range of the x axis under 4 meters.

FIG. 6A shows a cross-section view of monostatic millimeter-wave radar 504, according to an embodiment of the present invention. Millimeter-wave radar 504 includes die 608, balls 606, high frequency laminate 604 and antenna 602. Millimeter-wave radar 504 may be implemented, for example, as described in U.S. Pat. No. 9,935,065, filed on Dec. 21, 2016, in U.S. Patent Publication No. 2016/0178730, filed on Nov. 30, 2015, and in U.S. Patent Publication No. 2018/0074173, filed on Nov. 30, 2015, the contents of which are incorporated herein by reference in their entirety.

As shown in FIG. 6A, millimeter-wave radar 504 is implemented in monostatic configuration, in which the same antenna 602 is integrated in the same package, and is used for the transmitter (TX) module and for the receiver (RX) module. Implementing millimeter-wave radar 504 in monostatic configuration has the advantage of having a small footprint (e.g., in a device or system).

Antenna 602 is coupled to die 608, for instance, using conductive pillar 603. In some embodiments conductive pillar 603 is part of antenna 602 and is made with the same material as antenna 602. In other embodiments, the antenna may be remotely fed, for instance, through electromagnetic coupling.

High frequency laminate may be, for example, RO 4350 laminate from Rogers Corporation, Megtron 6 or 7 laminates from Panasonic, HL972 or HL 872 laminates from Mitsubishi. Other high-speed laminates may also be used.

Balls 606 are used to connect die 608 with external circuits. Some embodiments may implement pads instead of balls. Other connectors may also be used.

Die 608 includes a millimeter-wave radar sensor circuit (not shown). The millimeter-wave radar sensor circuit may

transmit and receive signals in the GHz range via antenna 602. For example, some embodiments may transmit and receive signals such as chirps in a band allocated around frequencies such as 95 GHz, 120 GHz, 140 GHz, and/or 240 GHz and/or other frequencies between about 95 GHz and about 240 GHz range. Other embodiments may transmit and receive signals such as chirps in the 20 GHz to 122 GHz range. Yet other embodiments may transmit and receive signals, such as chirps with frequencies above 240 GHz. Other frequencies and frequency ranges are also possible.

In some embodiments, the millimeter-wave radar sensor circuit process the echo signals received by using band-pass filter (BPFs), low-pass filter (LPFs), mixers, low-noise amplifiers (LNAs), and intermediate frequency (IF) amplifiers in ways known in the art. The echo signals are then digitized using one or more analog-to-digital converters (ADCs) for further processing. Other implementations are also possible.

Millimeter-wave radar 504 is capable of detecting the presence of objects (e.g., humans) in field-of-view 614 (i.e., the beam of millimeter-wave radar 504). As shown in FIG. 6A, the area of object detection varies based on the distance between the object and antenna 602. As shown, the diameter of coverage of field-of-view 614 may increase with height. For example, diameter d_1 , which corresponds to range (distance) r_1 to antenna 602, is smaller than diameter d_2 , which corresponds to range r_2 to antenna 602, where distance r_2 is larger than distance r_1 . The area of coverage is based on angle β , which may be, for example, between 45° and 50°. Angles smaller than 45° or bigger than 50° may also be used.

In some embodiments, millimeter-wave radar 504 determines the distance to the detected objects using range transformations, such as range FFT.

FIGS. 6B and 6C show a top view and a bottom view, respectively, of millimeter-wave radar 504, according to an embodiment of the present invention. FIG. 6D shows a schematic diagram of millimeter-wave radar 504, according to an embodiment of the present invention.

As shown, millimeter-wave radar 504 includes die 608 and antenna 602. Die 608 includes millimeter-wave radar sensor circuit 609, controller 618, and interface circuit 624. Millimeter-wave radar sensor circuit 609 includes front-end RF circuit 614, and mixed signal circuit 616. Controller 618 includes digital block 620 and signal processing block 622.

RF circuit 614 is configured to transmit and receive radar signals (e.g., chirps). RF circuit 614 includes transmitter circuit 610, receiver circuit 612. RF circuit 614 is implemented in a monostatic configuration.

Transmitter circuit 610 and receiver circuit 612 may be implemented in any way known in the art. Mixed signal circuit 616 is configured to control RF circuit 614 to transmit signals (e.g., chirps), and to receive the echo signal. Mixed signal circuit 616 is also configured to translate the RF signals into digital signals that are then transmitted to controller 618.

Mixed signal circuit 616 may be implemented in any way known in the art. For example, in some embodiments, mixed signal circuit 616 includes one or more band-pass filters (BPFs), low-pass filters (LPFs), mixers, low-noise amplifier (LNA), intermediate frequency (IF) amplifiers, phase-locked loops (PLLs) and analog-to-digital converters (ADCs).

Controller 618 is configured to process the signals received from millimeter-wave radar sensor circuit 609 and transmit it to an external processor (not shown in FIG. 6D). Controller 618 may be implemented in any way known in

the art, such as a general purpose controller or processor, ASIC, or any other implementation. Controller **618** typically includes digital block **620** for general control purposes (e.g., controlling millimeter-wave radar sensor circuit **609** and interface circuit **624**) and a signal processing block **622** for processing the signals received from millimeter-wave radar sensor circuit **609**. Digital block **620** may include a finite state machine (FSM).

Signal processing block **622** may be implemented with a digital signal processor (DSP). In some embodiments, signal processing block **622** implements a portion or all of processor **204**. In other embodiments, signal processing block **622** is not implemented and, instead, the raw data received from millimeter-wave radar sensor circuit **609** is sent to an external processor (not shown) for further processing. In some embodiments, millimeter-wave radar sensor circuit **609** may be implemented as an FMCW sensor.

Interface circuit **624** is configured to transmit data from controller **618** to the external processor (not shown). Interface **324** may be implemented, for example, using LPWAN (such as SigFox, NBIoT or LoRa). In some embodiments, interface **624** is implemented using wired communication protocols, such as serial peripheral interface (SPI), integrated circuit I²C, inter-IC source (I2S) or others. Other communication protocols, including other low data rate communication protocols, may be used.

Since radar signals of millimeter-wave radars, such as millimeter-wave radars **302**, **404**, and **504**, can penetrate materials such as plastic, it is possible to embed a millimeter-wave radar into a light fixture, such as a LED downlight having a plastic cover. For example, FIGS. 7A and 7B show a perspective view and a top view, respectively, of LED downlight **700** with an embedded millimeter-wave radar, according to an embodiment of the present invention. FIGS. 7C and 7D show schematic diagrams of a top view and a cross-sectional view, respectively, of LED downlight **700**, according to an embodiment of the present invention.

LED downlight **700** includes housing **702** and enclosure **708**. Housing **702** includes a plurality of LEDs **704**, and printed circuit board (PCB) **706**. PCB **706** includes a millimeter-wave radar, such as monostatic millimeter-wave radar **504**. Enclosure **708** includes power connector **712**, and supporting circuitry. Optional marker **710** may be included in housing **702** as a reference for direction (e.g., to be facing an entrance), as explained below.

As illustrated in FIGS. 7A-7D, LEDs **704** surround PCB **706**. As shown, such arrangement allows embedding PCB **706** without obstructing the light generated by LEDs **704**.

In some embodiments, PCB **706** may include monostatic millimeter-wave radar **504**. In such embodiments, and as shown in FIG. 7D, PCB **706** is disposed over the horizontal plane at an angle α . Such arrangement allows for achieving the angle α illustrated in FIG. 5A.

Marker **710** is optionally included to indicate the direction of field-of-view **614**. Such marker is advantageous, e.g., when installing LED downlight (e.g., at an entrance), such that the millimeter-wave radar determines which direction corresponds to humans entering the entrance and which direction corresponds to humans exiting the entrance.

Housing **702** is implemented with metal. Cover **711** is typically implemented with plastic. In some embodiments, cover **711** may be implemented with glass.

Since radar signals of millimeter-wave radars can penetrate the materials of cover **702**, millimeter-wave radars, such as millimeter-wave radars **300**, **404** and **504** are capable

of operating (e.g., detecting humans and humans' movements) while embedded inside housing **702** and covered by cover **711**.

Advantages of some embodiments include that a millimeter-wave radar may be embedded inside the housing of a light fixture without impact on aesthetics (e.g., the millimeter-wave radar may be invisible to a human crossing the plane). In some embodiments, a conventional housing of a light fixture has an embedded millimeter-wave radar without altering the housing shape. It is thus possible to implement, e.g., human detection and tracking functions, using existing light fixture designs. By embedding a millimeter-wave radar inside the light fixture, it is possible to perform functions such as human detection and tracking by using existing infrastructure and without using dedicated sensors that are installed separately. Some embodiments may exhibit an associated price/cost benefit for the user as a result of avoiding costly redesigns of existing infrastructure (e.g., existing building infrastructure) or installation of additional dedicated hardware.

FIG. 7E shows an electrical schematic of circuit **720** of LED downlight **700**, according to an embodiment of the present invention. Portions of circuit **720** may be implemented, for example, in enclosure **708** and PCB **706**. Circuit **720** includes power converter **724**, controller **730**, LED driver **726**, millimeter-wave radar **504**, wireless module **734**, and LED panel **728**.

During normal operation, circuit **720** is connected to AC power source **722** using connector **712**. Power converter **724** converts the AC power provided by AC power source **722** into DC power. The DC power provided by power converter **724** is provided to controller **730** and millimeter-wave radar **504**. LED driver receives power from AC power source **722** and drives LED panel **728** based on instructions received from controller **730**. Millimeter-wave radar **504** performs human detection, such as described with respect to FIG. 5A, and transmits the associated data to controller **730**. Controller **730** receives data from millimeter-wave radar **504** (e.g., human detection data), LED driver **726** (e.g., LED status) and/or power converter **724** (e.g., telemetry data) and transmits such data to wireless module **734**. Wireless module **734** transmits the data from controller **730** to wireless server **736**. Wireless module **734** may also receive data from wireless server **736** (e.g., instructions for dimming LEDs **704**) and transmits it to controller **730**.

Wireless module **734** may communicate with wireless server **736** using LPWAN protocols, such as SigFox, NBIoT or LoRa. Using an LPWAN protocol has the advantage of being capable of operating without user configuration. For example, in some embodiments, wireless module **734** is preconfigured during manufacturing. When a user installs LED downlight **700**, e.g., in a ceiling near an entrance of a shopping mall, and provides power (e.g., using connector **712**), wireless module **734** automatically establishes (e.g., anonymously) communication with wireless server **736**.

Modern LPWAN protocols, such as SigFox, NBIoT or LoRa, use a narrow-band sub-gigahertz frequency communication channel, which are more effective in penetrating, e.g., metal ceilings of modern buildings compared with existing gigahertz communication channels, such as Wifi or Bluetooth. Using modern LPWAN protocols for the wireless communication, therefore, also has the advantage of allowing effective wireless communications in modern buildings, e.g., that have metal ceilings.

AC power source **722** may be, for example, a 120 V_{rms}, 60 Hz power source or a 230 V_{rms}, 50 Hz. Other AC power sources may be used.

Power converter **724** may be implemented in any way known in the art. For example, power converter **724** may be implemented as a switched mode power supply (SMPS), for example. In some embodiments, power converter **724** may provide different voltage rails for different devices, such as a 3.3V rail for controller **730** and a 5 V rail for millimeter-wave radar **504**. In some embodiments, power converter **724** may include a charger circuit for charging a battery (not shown) configured to provide power if AC power source **722** becomes temporarily unavailable.

LED driver **726** may be implemented in any way known in the art. For example, LED driver **726** may be implemented as a current controlled SMPS. In some embodiments, LED driver **726** is capable of dimming LEDs **704**. In some embodiments, LED driver **726** receives DC power instead of AC power.

Controller **730** may receive data from millimeter-wave radar **504**, power converter **724** and/or LED driver **726**. For example, controller **730** may receive data associated with human detection (e.g., humans entering or exiting an entrance, count of humans over a period of time, etc.) from millimeter-wave radar **504**. In some embodiments, controller **730** may receive data associated with status of the grid (e.g., voltage and frequency of AC power provided by AC power source **722**) from power converter **724** and/or LED driver **726**.

In some embodiments, LED driver **726** may monitor parameters such as input voltage, power consumption of LED panel **728**, etc. Controller **730** may receive data of the monitored parameters, e.g., real time, from LED driver **726**. In some embodiments, controller **730** may estimate characteristics of LED panel **728** or one or more LEDs **704**, such as LEDs failures or end-of-life warnings, lumens produced, etc., and transmit such information to wireless server **736**. In some embodiments, controller **730** may transmit the data received from LED driver **726** to wireless server **736** and wireless server **736** may estimate parameters such as LEDs failures or end-of-life warnings, lumens produced, etc.

In some embodiments, controller **730** may receive data from external sources, such as other sensors (e.g., luminosity sensors, humidity sensors, temperature sensors, etc.) that are coupled to LED downlight **700** either directly or, in some embodiments, wirelessly. For example, in some embodiments, LED downlight **700** includes an embedded temperature sensor (not shown) to monitor the temperature of LEDs **704**. The temperature of LEDs **704** may be indicative, for example, of the luminosity produced by LEDs **704**.

Controller **730** may receive data and/or instructions from wireless server **736** and may use such information to dynamically adjust parameters of, e.g., LED driver **726**, and millimeter-wave radar **504**. For example, in some embodiments, controller **730** may receive from wireless server **736** instructions to dim (i.e., reduce luminosity), brighten (i.e., increase luminosity), turn on, or turn off, one or more LEDs **704**. As another non-limiting example, controller **730** may receive from wireless server **736** instructions to dynamically change a data collection method or processing technique of millimeter-wave radar **504**. In some embodiments, controller **730** may receive from wireless server **736** recalibration instructions dynamically to optimize for, e.g., operation at a particular height. Controller **730** may control, e.g., LED driver **726** and millimeter-wave radar **504** according to the instructions received from wireless server **736**.

In some embodiments, controller **730** may dim, brighten, turn on, or turn off one or more LEDs **704** based on data from millimeter-wave radar **504** and without relying on

instruction from wireless server **736**. For example, in some embodiments, such as in a store having a single entrance, controller **730** may dim LEDs **704** when the number of humans inside the (e.g., the store), e.g., as determined by subtracting the number of humans exiting the location from the number of humans entering the location, is below a threshold, such as 5 humans. Controller **730** may brighten LEDs **704** when the number of humans inside the location increases above, e.g., 10 humans.

Controller **730** communicates with millimeter-wave radar **504**, LED driver **726** and wireless module **734** using communication protocols known in the art. For example, some embodiments implement wired communication protocols, such as SPI, I²C, I2S, Universal asynchronous receiver-transmitter (UART), or others. Some embodiments, implement wireless communication protocols for such communication, such as Bluetooth, WiFi, or others.

Controller **730** may be implemented as a general purpose processor, controller or digital signal processor (DSP). In some embodiments, controller **730** may be implemented as a custom application specific integrated circuit (ASIC). In some embodiments, controller **730** includes a plurality of processors, each having one or more processing cores. In other embodiments, controller **730** includes a single processor having one or more processing cores.

In some embodiments, controller **730** includes controller **618**. In other embodiments, controller **618** is implemented separately from controller **730**.

PCB **706** includes a millimeter-wave radar, such as millimeter-wave radars **404** or **504**, such as shown in FIG. 6A-6D. In some embodiments, PCB **706** includes a portion of the millimeter-wave radar, such as the transmitting/receiving antenna, while other components of the millimeter-wave radar are implemented elsewhere. PCB **706** may be implemented in any way known in the art. For example, PCB **706** may be implemented as a flexible PCB or as a non-flexible PCB. PCB **706** may be implemented with a combination of low dielectric constant materials and FR4. In some embodiments, PCB **706** may be implemented with purely FR4 if carefully designed. Other types of materials may also be used.

Wireless server **736** may be a remote server. For example, in some embodiments, wireless server **736** is a server located miles away from LED downlight **700**, such as in a centralized server farm. In some embodiments, wireless server **736** is connected to the cloud and may be accessed, e.g., via the Internet, by a personal computer, smartphone, tablet, or other electronic device.

Advantages of some embodiments include that a millimeter-wave radar embedded in a light fixture may operate by using existing power source connections. In some embodiments, the wireless communication channel used to transfer information from the millimeter-wave radar (or other source of data inside the light fixture) to a wireless server or to provide instructions from a wireless server to control parameters of the light fixture is advantageously preconfigured and operates without human intervention. Thus, some embodiments advantageously exhibit plug-and-play capability. In other words, in some embodiments, a user installs the light fixture having an embedded millimeter-wave radar by connecting the light fixture to AC power and, after power-up, the user has access to data collected by the millimeter-wave radar (e.g., number of humans entering or exiting an entrance) by using an electronic device (e.g., a smartphone, computer, or tablet) without additional configuration of the light fixture. In some embodiments, the plug-and-play capa-

bility extends to allow the user control of parameters of the light fixture, such as, for example, changing the brightness level of one or more LEDs.

FIGS. 7F and 7G show a perspective view of portions of the circuitry inside enclosure **708**, according to an embodiment of the present invention. As shown in FIG. 7F, enclosure **708** includes PCB **738**. PCB **738** includes power converter **724**, controller **730** (not shown) and wireless module **734** (not shown). As shown in FIG. 7G, enclosure **708** also includes PCB **740** that includes LED driver **726**. Some embodiments may implement the circuitry inside enclosure **708** in a different manner (e.g., using a single PCB or more than two PCBs. For example, some embodiments may implement some of the circuitry shown in FIGS. 7F and 7G inside housing **702** (such as in PCB **706**).

LED downlight **700** includes monostatic millimeter-wave radar **504**. In some embodiments, LED downlight **700** may be implemented with other types of millimeter-wave radars, such as, for example, millimeter-wave radar **404**.

FIG. **8A** shows a top view of a schematic diagram of building **802** that includes system **800**, according to an embodiment of the present invention. System **800** includes a plurality of LED downlights **700** and wireless server **836**.

Building **802** includes a plurality of entrances **804**. A LED downlight **700** is disposed in front of each entrance **804**. As shown in FIG. **8A**, each LED downlight **700** is oriented towards respective entrances **804**, as indicated by optional market **710**.

Wireless server **836** communicates with each LED downlight **700** using an LPWAN protocol, such as SigFox, NB-IoT or LoRa. Upon installation of LED downlight **700** (e.g., providing power to LED downlight **700**), each LED downlight **700** communicates with wireless server **836**. For example, in some embodiments, each LED downlight periodically transmits a log that includes each human detected, a time stamp of the human detected, and the direction of the human detected (e.g., entering or exiting the respective entrance).

In some embodiments, wireless server **836** stores, for each sampled point (e.g., for each log record), the device id (to identify which LED downlight **700** is associated with the data), a time stamp (e.g., day and time to identify period of detection), number of humans approaching the entrance (e.g., entering building **802**), number of humans leaving the entrance (e.g., exiting building **802**), the entrance id (to identify the entrance, which may be derived from the device id), and a description field (e.g., such as “mini bus entrance,” “coffee shop entrance,” etc.).

In some embodiments, each LED downlight **700** periodically reports to wireless server **836** how many humans have entered/exited during an associated period. For example, every, e.g., 5 minutes, each LED downlight **700** reports how many humans entered and exited during such 5 minutes. In other embodiments, each LED downlight **700** asynchronously reports to wireless server **836** that a human has been detected entering/exiting the entrance.

Wireless server **836** performs data management and stores information received from each LED downlight **700**, for example, in database **812**. Wireless server **836** may also aggregate and process information from each LED downlight **700** and make it accessible to external electronic devices **810**. For example, wireless server may compute how many humans are inside building **802** by adding all humans entering building **802** and subtracting all humans exiting building **802**.

Wireless server **836** may also determine peak hours of operation (hours with the most number of humans), average number of humans, and other metrics based on data gathered from LED downlights **700**.

Wireless server **836** may also determine, for example, expected life remaining for each LED downlight **700**. For example, the power consumption of a LED light goes down near end of life. LED downlight **700** may monitor the power consumption of its associated LEDs **704** to determine LEDs **704** end-of life. Such information (e.g., power consumption of LEDs **704** and/or end-of-life estimates) may be transmitted to wireless server **836**, which then may alert a user of potential lighting disruptions. Wireless server **836** may also automatically compensate for a decrease in luminosity of a near-end-of-life LED downlight by adjusting the brightness of such LED.

Wireless server **836** may also receive telemetry information (e.g., voltage and frequency of AC power source **722**) from each LED downlight **700** and make such data available to a user (e.g., via electronic devices **810**). Telemetry information may also be used, for example, to confirm that an action, such as turning on or off LED downlight **700**, was actually carried out. As another example, the telemetry information may be used to confirm which dimming level is currently in operation (e.g., based on power consumption of LED panel **728**).

Wireless server **836** may also control one or more LED downlights **700** based on the information from each LED downlight **700**, as well as from external devices **810**. For example, wireless server **836** may dim downlights **700** (by, e.g., instructing respective controllers **730** using the LPWAN) when the number of humans inside building **802** (or a portion of building **802**) is below a threshold (e.g., 20 humans). In some embodiments, electronic devices **810** may be used to control one or more LED downlights **700**. For example, a smartphone may be used to turn on, off, or change the brightness of one or more LED downlights **700**, e.g., using an app in a smartphone.

Electronic devices **810** include, for examples, personal computers, smartphones and tablets. Electronic devices **810** may communicate with wireless server **836**, for example, via the Internet. For example, electronic devices **810** may access data contained in wireless server **836** by logging into a website or using an app.

Wireless server **836** may, for example, use a cloud computing platform (such as Amazon Web Services, Microsoft Azure, and Google Cloud platform) to host data, and app and website servers to communicate with electronic devices **810**, e.g., via the Internet.

In some embodiments, wireless server **836** may store historic data of one or more LED downlights **700** and use such data to predict behavior and dynamically adjust LED downlights **700** based on such predictions. For example, data mining techniques may be used to predict peak human traffic on a location on a particular day (e.g., a typical rainy Saturday). Such prediction may be used to dynamically adjust the dimming of LED lights **700**. Such data and predictions may also be used, e.g., to calculate previous electricity usage of LED downlights **700** and to estimate projected electricity usage of LED downlights **700**. In some embodiments, artificial intelligence (AI) and machine learning algorithms may be used to optimize the dynamic control of LED lights **700**.

Building **802** may be, for example, a shopping mall, office building, or factory. Other buildings, such as a residential house, convention center, hotel, or other, may also be used. It is understood that system **800** may also be implemented

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outside a building. For example, in some embodiments, system **800** may be implemented in street lights to determine human traffic, e.g., in the sidewalks. The form factor and other characteristics may be adapted to be suitable for the particular application. For example, a street light may be designed to be weatherproof.

Advantages of some embodiments include that a building may implement a system such as system **800** by replacing existing downlights with downlights implementing with downlights according to embodiments of the present invention. Since LED downlights **700** operate by using a similar interface than conventional downlights (e.g., connector **712** receiving AC power from the grid) and incorporates the millimeter-wave radar inside housing **702**, some embodiments implement system **800** in a building without impact to aesthetic and without installing additional sensors or wiring. Some embodiments may reuse **100%** of existing infrastructure (e.g., wiring, ceiling pockets, etc.).

Using millimeter-wave radars inside downlights has the advantage of achieving a large coverage range (e.g., greater than 10 meters in the x-axis), in part, because radar signals from the millimeter-wave radar can penetrate plastic. The capability of being dynamically calibrated advantageously allows for installing the same downlight model at different heights and under different conditions without impacting performance.

Using millimeter-wave radars has the additional advantage of avoiding privacy issues associated with capturing and storing images of humans and performing facial recognition and other signal processing techniques with such images.

Embodiments using LPWAN protocols have additional advantages, such as avoiding wireless configuration steps when installing the downlights in the building. Using LPWAN also has the advantage of allowing effective wireless communications in modern buildings, e.g., buildings that have metal ceilings, because of LPWAN's usage of narrow-band sub-gigahertz frequency communication channels.

Using LPWAN protocols is possible, in part, thanks to the relatively low data usage and low complexity of system **800**. For example, it is possible to count human entering and exiting an entrance locally in the LED downlight without complex radar signal processing, thereby transmitting a small log file/data (e.g., containing time stamp and direction of detection) periodically. Alternatively, data of detected humans may be transmitted to the wireless server for remote processing (e.g., remotely determining direction of human movement or filtering out noise).

Additional advantages of some embodiments include that human counting and monitoring may be performed automatically twenty-four hours a day, seven days a week. For example, human counting may continue through the night and holidays without user intervention. Autonomous interaction between the wireless server and each of the downlights is advantageously possible. In other words, the wireless server may dynamically adjust the state of the downlights (e.g., on/off and dimming state) based on current human detection as well as historic data and estimated human traffic without user intervention.

In some embodiments, wireless server **836** may use artificial intelligence (AI) and machine learning algorithms to process data from LED downlights **700**. For example, in some embodiments, correlation between human detection of various LED downlights **700** may be used to estimate human flow inside building **802**.

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Although FIG. **8A** shows a single LED downlight **700** per entrance, more than one LED downlights **700** may be used per entrance. Some embodiments may also use LED downlights **700** inside the building (not only at the entrances) to, e.g., monitor human flow inside the building.

Some embodiments may be adapted to detect other objects, such as cars. For example, some embodiments may detect the number of cars moving, their associated speeds, and car congestion using system **800** implemented in street lights and/or traffic lights. As another example, some embodiments may determine, e.g., using millimeter-wave radars embedded in traffic lights, the number of cars crossing a first intersection in a first direction, and a second intersection subsequent the first intersection in the first direction. Machine learning and AI algorithms may then be used to adjust the timing of the red and green lights of the first and second traffic lights to, e.g., reduce car congestion.

FIG. **8B** shows graphical user interface (GUI) **820** that may be used to access wireless server **836**, according to an embodiment of the present invention. GUI **820** is a non-limiting example of a possible GUI to interface with wireless server **836**. Other user interfaces, including other GUIs, are also possible. GUI **820** includes building diagram **824**, relative building traffic diagram **826**, peak hour diagram **832**, major exit gate diagram **830**, energy savings diagram **822**, and total human traffic diagram **834**.

Building diagram **824** is a diagram of building **802** and displays in real time how many humans are entering and exiting building **802** on each entrance. For example, the bars next to the human symbol in each entrance may be used to indicate the number of humans entering building **802** (e.g., the bars towards the inside of building **802** with respect to the human symbol) and the number of humans exiting building **802** (e.g., the bars towards the entrance of building **802** with respect to the human symbol), with the number of bars being indicative of the number of humans entering/exiting building **802**. In some embodiments different colors may be used for the bars (e.g., red for entering and blue for exiting).

Relative building traffic diagram **826** displays the relative human traffic of each entrance **804** for, e.g., the current day (only 4 entrances are displayed in relative building traffic diagram **826** for clarity purposes). For example, the number of human symbols next to each entrance represents the relative number of humans entering and exiting a particular entrance with respect to other entrances. In the example shown, all entrances displayed have similar human traffic.

Peak hour diagram **832** displays in each hollow circle the relative number of human traffic per hour in the respective entrance, e.g., using different shades of a color, for the current day (only 4 entrances are displayed in this example for clarity purposes). For example, each hollow circle is divided in 24 equal portions, and a darker color in a particular portion of the 24 portions indicates higher human traffic than lighter colors. In the example shown, the south entrance has higher traffic between 1 m and 10 am and between 7 pm and 11 pm. The peak human traffic for the east entrance is at 1 pm and at 11 pm.

Major exit gate diagram **830** displays the percentage of exiting traffic per entrance for the current day (only 4 entrances are displayed in this example for clarity purposes). In the current example, for the current day, 20.3% of humans exited building **802** using the east entrance, 36.2% of humans exited building **802** using the south entrance, 36.2% of humans exited building **802** using the north-east entrance, and 9.1% of humans exited building **802** using the west entrance.

Total human traffic diagram **834** displays total number of humans entering and exiting all entrances for the current day. In the current example, 1000 humans entered or exited an entrance of building **802** during the current day.

Energy savings diagram **822** displays how much electricity has been saved during the current day as a result of dimming LED downlights **700** when human traffic is low. For example, the number of light-bulbs in energy savings diagram **822** is indicative of the number of kW saves as a result of dimming one or more LED downlights **700**.

Some embodiments may integrate system **800** with other types of sensors. For example, FIG. **9** shows a top view of a schematic diagram of building **902** that includes system **900**, according to an embodiment of the present invention. System **900** includes a plurality of LED downlights **700**, inside temperature sensor **910**, outside temperature sensor **912**, pressure sensors **906**, and wireless server **936**.

System **900** may operate in a similar manner than system **800**. System **900**, however, is capable of interacting with additional sensors, such as temperature sensors **910** and **912**, and pressure sensors **906**. System **900**, therefore, may advantageously use information from such additional sensors to dynamically adjust LED downlights **700** and external systems, such as air conditioner **916**, and security system **914**. For example, some embodiments, may use the current and/or estimated number of humans inside building **902**, current and/or estimated building temperature (e.g., from inside temperature sensor **910**) and current and/or external temperature (e.g., from outside temperature sensor **912**) to control air conditioner **916**.

As another non-limiting example, some embodiments may use human detection capabilities of downlights **700** together with pressure sensors **906** (located in windows **904**) to trigger security system (e.g., alarm siren, police call, etc.). Security system may also dynamically adjust the brightness of LED downlights **700** based on, e.g., intrusion detections. For example, in some embodiments, pressure sensor **906** may report to wireless server **936** that a window was broken, wireless server may then turn on all LED downlights **700** at full brightness.

In some embodiments air conditioner **916** and alarm system **914** communicate with wireless server **936** via the Internet. In other embodiments, air conditioner **916** and security system **914** communicated with wireless server **936** using LPWAN protocols. Other communication channels may also be used.

In some embodiments, wireless server **936** may implement data mining, AI and machine learning algorithms to data associated with external sensors (in addition to data from LED downlights **700**). Other tasks, such as estimation of electricity consumption associated with the air conditioner, control of air conditioner, alarm system and control of other systems, are also possible.

Advantages of some embodiments include enabling various systems (e.g., human detection, lighting, air conditioner, and security/alarm systems, among others) to interact seamlessly without human intervention. In some embodiments, the wireless server advantageously serves as a gateway for smart buildings and allows for the addition of additional sensors and systems.

FIG. **10** shows LED downlights **700** installed on a ceiling, according to an embodiment of the present invention. As shown, two LED downlights **700** are used in a security entrance inside a building (which may be inside an office building or an airport, for example). As shown, LED downlights **700** are aesthetically similar to other LED downlights in the entrance.

Example embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

Example 1. A downlight including: a plurality of light emitting diodes (LEDs) disposed in a housing of the downlight; and a millimeter-wave radar including: an antenna disposed in the housing, a controller configured to: detect a presence of a human in a field-of-view of the millimeter-wave radar, determine a direction of movement of the detected human, and produce log data based on the direction of movement of the detected human, and a wireless module configured to transmit the log data to a wireless server.

Example 2. The downlight of example 1, where the field-of-view of the millimeter-wave radar includes a centerline that has an angle with respect to a vertical axis that is different than 0° .

Example 3. The downlight of one of examples 1 or 2, where the antenna is disposed in a printed circuit board (PCB), where the plurality of LEDs are disposed around the PCB.

Example 4. The downlight of one of examples 1 to 3, where the antenna is integrated in an integrated circuit that includes the millimeter-wave radar, and where the angle is between 5° and 15° .

Example 5. The downlight of one of examples 1 to 4, where the antenna is disposed in a printed circuit board (PCB), where the plurality of LEDs are disposed around the PCB, and where the PCB is in a plane that has an angle with respect to a horizontal axis that is different than 0° .

Example 6. The downlight of one of examples 1 to 5, where the angle is between 5° and 15° .

Example 7. The downlight of one of examples 1 to 6, where the field-of-view of the millimeter-wave radar determines a plane, and where the controller is configured to count a number of humans crossing the plane and to transmit the count to the wireless server using the wireless module as part of the log data.

Example 8. The downlight of one of examples 1 to 7, where the field-of-view of the millimeter-wave radar defines a plane, where the controller is configured to determine a first direction of movement of the detected human when the human crosses the plane from a first side to a second side, and determine a second direction of movement of the detected human when the human crosses the plane from the second side to the first side, and where the log data includes a direction of movement of the detected human.

Example 9. The downlight of one of examples 1 to 8, and where the controller is further configured to, during a time interval, determine a first count of humans crossing the plane from the first side of the plane to the second side of the plane, determine a second count of humans crossing the plane from the second side of the plane to the first side of the plane, and transmit the first and second counts to the wireless server using the wireless module as part of the log data.

Example 10. The downlight of one of examples 1 to 9, where the controller is configured to periodically determine the first and second counts and periodically transmit the first and second counts to the wireless server.

Example 11. The downlight of one of examples 1 to 10, where the controller is further configured to determine a range component of a location of the detected human, and where the controller is configured to determine the direction of movement of the detected human based on the range component.

Example 12. The downlight of one of examples 1 to 11, further including: a power converter configured to receive an

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AC supply voltage via a power connector; and a LED driver coupled to the plurality of LEDs, where the controller is configured to: receive data from the wireless module, and control the LED driver based on data received from the wireless module.

Example 13. The downlight of one of examples 1 to 12, where the controller is configured to adjust a luminosity of the plurality of LEDs by controlling the LED driver based on data received from the wireless module.

Example 14. The downlight of one of examples 1 to 13, where the LED driver is configured to determine a state of the AC supply voltage, and where the controller is configured to transmit the state of the AC supply voltage to the wireless server using the wireless module.

Example 15. The downlight of one of examples 1 to 14, where the wireless module is preconfigured such that the wireless module is configured to communicate with the wireless server when the power converter receives the AC supply voltage without user intervention.

Example 16. The downlight of one of examples 1 to 15, where the wireless module is configured to transmit data using a low-power wide-area network (LPWAN) protocol.

Example 17. The downlight of one of examples 1 to 16, where the log data includes the direction of movement of the human and a time stamp of the detected presence of the human.

Example 18. The downlight of one of examples 1 to 17, where the downlight is configured to be installed at a ceiling having a first height to a floor, where the controller is configured to calibrate the millimeter-wave radar to operate at the first height.

Example 19. The downlight of one of examples 1 to 18, where the controller is further configured to determine a frequency change between a signal transmitted from the millimeter-wave radar and a signal received by the millimeter-wave radar, and where the controller is configured to determine the direction of movement of the detected human based on the frequency change.

Example 20. The downlight of one of examples 1 to 19, where the millimeter-wave radar is a monostatic millimeter-wave radar.

Example 21. The downlight of one of examples 1 to 20, where the millimeter-wave radar is configured to determine a location of the detected human using elevation, azimuth, and range components.

Example 22. A method including: detecting a presence of a human in a field-of-view of a millimeter-wave radar embedded in a housing of a downlight; determining a direction of movement of the detected human; producing log data based on the direction of movement of the detected human; and transmitting the log data to a server using a wireless transmission channel.

Example 23. The method of example 22, further including: receiving log data from a plurality of downlights, the plurality of downlights including the downlight; determining a number of humans inside a building based on the log data from the plurality of downlights; and controlling a brightness of light produced by the plurality of downlights based on the determined number of humans inside the building.

Example 24. The method of one of examples 22 or 23, further including dimming the downlights when the determined number of humans inside the building is below a threshold.

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Example 25. The method of one of examples 22 to 24, further including controlling an operation of an air conditioner of the building based on the determined number of humans inside the building.

Example 26. The method of one of examples 22 to 25, further including: receiving data from a pressure sensor located at a window or door of the building; and controlling a brightness of light produced by the downlight based on the data received from the pressure sensor.

Example 27. A method including: determining whether humans are entering or exiting a building using millimeter-wave radars embedded in respective housings of respective downlights of a plurality of downlights located at entrances of the building; calculating a first number of humans entering the building during a first period of time; calculating a second number of humans exiting the building during the first period of time; and controlling a brightness of light produced by the plurality of downlights based on the first and second number of humans.

Example 28. The method of example 27, further including: storing the first and second numbers in a database; and estimating a number of humans entering or exiting the building based on data from the database.

Example 29. The method of one of examples 27 or 28, further including: storing the first and second numbers in a database; and determining a peak time of human traffic based on data from the database.

Example 30. The method of one of examples 27 to 29, further including: accessing the database with an electronic device via the Internet; and displaying the peak time in the electronic device.

While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

What is claimed is:

1. A method comprising:

detecting a presence of a human in a field-of-view of a millimeter-wave radar embedded in a housing of a downlight;

determining a direction of movement of the detected human using the millimeter-wave radar;

producing log data based on the direction of movement of the detected human; and

transmitting the log data to a server using a wireless transmission channel.

2. The method of claim 1, further comprising:

receiving log data from a plurality of downlights, the plurality of downlights comprising the downlight;

determining a number of humans inside a building based on the log data from the plurality of downlights; and

controlling a brightness of light produced by the plurality of downlights based on the determined number of humans inside the building.

3. The method of claim 2, further comprising dimming the downlights when the determined number of humans inside the building drops below a threshold.

4. The method of claim 2, further comprising controlling an operation of an air conditioner of the building based on the determined number of humans inside the building.

5. The method of claim 1, further comprising:

receiving data from a pressure sensor located at a window or door of a building; and

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controlling a brightness of light produced by the downlight based on the data received from the pressure sensor.

6. The method of claim 1, wherein the wireless transmission channel is based on a low-power wide-area network (LPWAN) protocol.

7. The method of claim 1, wherein the millimeter-wave radar is implemented in a monostatic configuration.

8. The method of claim 1, wherein the millimeter-wave radar is implemented in a bistatic configuration.

9. The method of claim 1, wherein detecting the presence of a human comprises:

determining a range to an object in the field-of-view of the millimeter-wave radar; and

detecting the presence of the human when the range to the object increases by an average height of a human.

10. The method of claim 9, wherein determining the direction of movement of the detected human comprises:

when the range to the object increases by a first value smaller than the average height of the human, determining the direction of movement as a first direction; and

when the range to the object decreases by a second value smaller than the average height of the human, determining the direction of movement as a second direction, the second direction being opposite to the first direction.

11. The method of claim 10, further comprising determining whether the detected human is entering or exiting a building based on the determined direction.

12. The method of claim 9, further comprising determining an age of the detected human based on the range.

13. The method of claim 1, wherein detecting the presence of a human comprises:

transmitting radar signals with the millimeter-wave radar; receiving reflected radar signals with the millimeter-wave radar;

determining a Doppler frequency change between transmitted radar signals and the received reflected radar signals; and

detecting the presence of the human when the Doppler frequency change is higher than a predetermined threshold.

14. The method of claim 13, wherein determining the direction of movement of the detected human comprises:

when the Doppler frequency is higher than a transmission frequency of the transmission radar signals, determining the direction of movement as a first direction; and when the Doppler frequency is lower than the transmission frequency, determining the direction of movement as a second direction, the second direction being opposite to the first direction.

15. The method of claim 1, wherein the millimeter-wave radar performs the determination of the direction of movement of the detected human.

16. The method of claim 1, further comprising:

transmitting a plurality of radar signals at different strengths using the millimeter-wave radar without having a human in the field-of-view of the millimeter-wave radar;

receiving first reflected radar signals associated with the plurality of radar signals transmitted at different strengths; and

selecting one of the different strengths based on the first reflected radar signals.

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17. The method of claim 16, further comprising:

transmitting radar signals at the selected one strength using the millimeter-wave radar with a human in the field-of-view of the millimeter-wave radar;

receiving second reflected radar signals associated with the radar signals transmitted at the selected one strength; and

determining a threshold for determining whether the presence of the human is detected based on the second reflected radar signals.

18. A method comprising:

receiving log data from a plurality of downlights via a wireless transmission channel, wherein the log data comprises data based on a direction of movement of a human detected by a millimeter-wave radar embedded within a housing of at least one downlight of the plurality of downlights;

determining a number of humans inside a building based on the log data from the plurality of downlights; and controlling a brightness of light produced by the plurality of downlights based on the determined number of humans inside the building.

19. The method of claim 18, further comprising dimming the downlights when the determined number of humans inside the building drops below a threshold.

20. The method of claim 19, further comprising controlling an operation of an air conditioner of the building based on the determined number of humans inside the building.

21. The method of claim 18, wherein the wireless transmission channel is based on a low-power wide-area network (LPWAN) protocol.

22. A downlight comprising:

a housing;

an illumination device configured to produce visual light; a millimeter-wave radar device embedded in the housing of the downlight;

a wireless communication device;

a controller configured to:

detect a presence of a human in a field-of-view of the millimeter-wave radar device embedded in the housing of the downlight using the millimeter-wave radar device;

determine a direction of movement of the detected human using the millimeter-wave radar device;

produce log data based on the direction of movement of the detected human; and

transmit the log data to a server via a wireless transmission channel using the wireless communication device.

23. The downlight of claim 22, wherein the controller is further configured to:

determine a range to an object in the field-of-view of the millimeter-wave radar device; and

detect the presence of the human when the range to the object increases by an average height of a human.

24. The downlight of claim 23, wherein the controller is further configured to:

when the range to the object increases by a first value smaller than the average height of the human, determine the direction of movement as a first direction; and when the range to the object decreases by a second value smaller than the average height of the human, determine the direction of movement as a second direction, the second direction being opposite to the first direction.

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25. The downlight of claim **24**, wherein the controller is further configured to determine whether the detected human is entering or exiting a building based on the determined direction.

26. The downlight of claim **22**, wherein: 5
the millimeter-wave radar device is further configured to:
transmit a plurality of radar signals at different
strengths using the millimeter-wave radar without
having a human in the field-of-view of the millime-
ter-wave radar device, and 10
receive first reflected radar signals associated with the
plurality of radar signals transmitted at different
strengths; and
the controller is further configured to select one of the
different strengths based on the first reflected radar 15
signals.

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